

Growth Accounting with Time-Varying Input Intensities^{*}

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Abstract

Does the assumption of constant intermediate input shares affect measured total factor productivity (TFP)? This paper shows that it does, and that the effect is both systematic and quantitatively large. The standard growth accounting framework with intermediate goods assumes that the share of gross output devoted to intermediate inputs is fixed over time. We develop a framework that relaxes this assumption. Applying this framework to 14 advanced economies over 1995 to 2020, we show that allowing intermediate input intensities to vary over time can meaningfully shift measured sectoral and aggregate TFP, with the direction and magnitude depending on the country, sector, and time period. For example, by 2020, aggregate TFP in Luxembourg is approximately 26 percentage points higher under the constant-share approach, Germany's is nearly 13 percentage points higher, whereas Latvia's is about 8 percentage points lower. The gap differs systematically across goods-producing sectors and services due to their structurally distinct input intensities. These differences have direct implications for the literature on structural transformation, as measured sectoral productivity differentials, which drive labor reallocation in standard models, depend non-trivially on how intermediate input intensity is treated. As global value chains expand and production networks deepen, assuming fixed input shares becomes increasingly restrictive. Accounting for the evolution of production structure is therefore essential for accurate productivity measurement.

Keywords: Growth accounting, intermediate-input use, total factor productivity measurement, structural transformation, development heterogeneity.

JEL Classification: C43, C82, D24, E23, O11, O47

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1 Introduction

Accurate measurement of productivity is central to understanding long-run economic growth and structural transformation. Sectoral and aggregate productivity estimates are widely used to explain cross-country income differences (Caselli, 2005; Valentinyi, 2021), patterns of industrialization (Huneus and Rogerson, 2023; Sposi et al., 2025), and the reallocation of resources across agriculture, manufacturing, and services (Ngai and Pissarides, 2007; Herrendorf et al., 2014; Swiecki, 2017; Sáenz, 2022; Marcolino, 2022). Productivity is typically measured using either a value-added production function or a gross-output production function. When the gross-output framework is adopted, the analysis must explicitly account for intermediate inputs. However, most empirical studies assume that the intensity of intermediate input use remains constant over time. This assumption is restrictive, as the share of intermediate inputs in production can evolve substantially with the deepening of supply chains and changes in input–output linkages across sectors.¹ This paper develops a framework that allows the intensity of intermediate inputs to vary over time to measure sectoral and aggregate total factor productivity (TFP).

The distinction between value-added and gross-output TFP measures carries significant implications for the analysis of economies characterized by inter-sectoral linkages through intermediate input use. While value-added approaches abstract from intermediate goods by netting them out of production, gross-output approaches explicitly incorporate them as productive inputs. Although these two frameworks are theoretically isomorphic, the TFP estimates they generate diverge in meaningful ways. As Ngai and Samaniego (2009) demonstrates, the relationship between value-added and gross-output TFP is non-trivial in multisector economies with intermediate goods, given that changes in the composition and use of intermediate inputs can alter the mapping between sectoral prices and measured TFP. This distinction becomes particularly consequential when TFP improvements occur in sectors that supply intermediate goods to downstream industries, since such gains propagate through input–output linkages and affect the costs and prices faced by sectors that rely on those inputs.

While recent multi-sectoral growth models incorporate intermediate input use, they nevertheless tend to rely on a simplifying assumption: the share of intermediate inputs in gross output is constant over time. This approach is methodologically convenient; however, it may potentially overlook an important dimension of changes in the production structure.² Using panel data for a broad set of countries, we document substantial and heterogeneous changes in intermediate input intensities across sectors and over time. Intermediate input shares do not follow a uniform pattern across economies. In some countries, they rise steadily, while in others they decline, and their trajectories differ markedly across agriculture, manufacturing, and services.

These evolutions of intermediate input intensities challenge the conventional assumption of

¹Input–output linkages refer to the intersectoral connections whereby the output of one sector is used as an intermediate input in the production of other sectors.

²A growing body of evidence documents that intermediate input intensities are far from stable, varying systematically over time (Choudhry, 2021; Gaggi et al., 2023; Avoumatsodo and Leunga Noukwe, 2024). The assumption of fixed intermediate shares thus risks biasing the measurement and interpretation of productivity dynamics.

fixed input shares and raise several open questions: How do such changes alter the measurement of productivity across countries and sectors? Do the potential measurement gaps from holding input intensities constant vary systematically across sectors, or are they uniform across economies? These questions remain largely unexplored, even as debates on the sources of productivity growth continue. This motivates the development of a growth accounting framework that explicitly incorporates evolving sectoral input intensities into the assessment of productivity.

Therefore, in this paper, we build a multi-sector model with time-varying intermediate input intensities to study how these variations affect productivity measurement. In our framework, representative firms in agriculture, manufacturing, and services produce gross output using capital, labor, and intermediate goods, with intermediate input intensities evolving over time. We calibrate the model and implement a growth accounting exercise to construct sectoral and aggregate TFP growth for a panel of countries under two alternative assumptions: constant versus time-varying intermediate input intensities. This approach allows us to evaluate whether holding input intensities fixed affects productivity dynamics and to examine how such effects differ across sectors and countries.

The findings reveal that across 14 advanced economies over the period 1995–2020, allowing intermediate input intensities to vary over time can meaningfully shift measured sectoral and aggregate TFP, with the direction and magnitude depending on the country, sector, and time period. For example, by 2020, aggregate TFP in Luxembourg is approximately 26 percentage points higher under the constant-share approach, Germany’s is nearly 13 percentage points higher, whereas Latvia’s is about 8 percentage points lower. The gap differs systematically across goods-producing sectors and services due to their structurally distinct input intensities.³

When intermediate intensities rise, fixing the share at a constant level attributes too large a fraction of output to value added, thereby overstating value-added productivity. A second source of measurement gap operates through aggregation. Gross-output productivity combines value-added productivity and intermediate-input efficiency using weights that depend on the intermediate share. When intermediate intensities change, these weights shift accordingly. Holding the share fixed therefore distorts the relative contribution of the two components to measured gross-output productivity, even when their underlying productivity levels are correctly measured.

This aggregation effect depends on the relative importance of intermediate input shares in gross output. In services, where value added constitutes the dominant component of production, an increase in input intensity tends to reduce measured gross-output TFP growth. By contrast, in agriculture and manufacturing, where production relies more heavily on intermediate inputs, increases in intermediate intensities tend to raise measured gross-output productivity. Because sectoral productivity differentials are a key driver of labor reallocation across sectors, these differences also matter for the interpretation of structural transformation.

Related Literature. This paper is related to the literature examining the implications of intermediate input use for measured TFP. A foundational contribution is [Hulten \(1978\)](#), who

³Although our quantitative analysis focuses on advanced economies due to data constraints, the paper’s methodological contribution, namely a tractable and national accounts consistent framework for measuring TFP with time varying input intensities, has broad applicability. Extending this framework to developing and emerging economies, where production structures evolve more rapidly, constitutes a promising avenue for future research.

shows that productivity shocks propagate through input–output linkages across sectors. [Jorgenson et al. \(1987\)](#) provides the canonical multi-sector implementation of this framework, constructing KLEMS-based productivity accounts for the U.S. economy with intermediate inputs treated as explicit productive inputs in each industry. Because sectors use each other’s output as intermediate inputs, productivity improvements in one sector can have amplified aggregate effects through production networks. [Acemoglu et al. \(2012\)](#) formalizes this propagation mechanism in a network setting, showing that the structure of input–output linkages determines how sectoral shocks translate into aggregate fluctuations. Subsequent work has further explored these mechanisms. [Ciccone \(2002\)](#) emphasizes how intermediate goods generate increasing returns at the aggregate level, while [Jones \(2011\)](#) shows that intermediate goods create a multiplier effect on productivity that depends on the intensity with which intermediates are used in production.

Another strand of the literature studies how the presence of intermediate goods affects productivity measurement and growth accounting. [Basu and Fernald \(2002\)](#) provides a general accounting framework that distinguishes aggregate productivity from aggregate technology in the presence of intermediate inputs and market distortions, deriving exact expressions that link value-added and gross-output productivity measures and showing that imperfect competition in intermediate-good-supplying sectors drives a wedge between the two that does not appear in standard value-added accounting. [Ngai and Samaniego \(2009\)](#) shows that accounting for intermediate goods can substantially reshape growth accounting results by altering the mapping between observed prices and productivity measures. In particular, they demonstrate that distinguishing between value-added and gross-output productivity can significantly change the measured contribution of investment-specific technical change to post-war U.S. growth.

Our work differs from these contributions in that they typically assume constant intermediate input intensities, whereas we allow input intensities to vary over time in line with the data and show that the resulting measurement gap between the two approaches is both systematic and quantitatively significant.

Most closely related to the present paper are [Choudhry \(2021\)](#), [Moro \(2012\)](#), and [Esfahani et al. \(2024\)](#). [Choudhry \(2021\)](#) shows that industries in India’s formal manufacturing sector with higher intermediate input intensity tend to exhibit lower productivity, focusing on a single country and a single sector. [Moro \(2012\)](#) introduces intermediates-biased technical change as an exogenous structural driver of the evolution of intermediate input use, and shows that this mechanism shapes measured aggregate TFP dynamics for Italy. [Esfahani et al. \(2024\)](#) employs a global growth accounting framework to decompose world GDP growth into parts driven by technology, labor, and capital, as well as by factor reallocation and markups. They find that world productivity growth is volatile from year to year and that this mainly reflects the reallocation of labor across country-industries. They also show that the contribution of country-industry level TFP growth to world TFP is relatively constant over time. Our study differs from these papers in three important respects.

First, we propose a new framework to measure TFP growth when intermediate input intensities evolve, along with a complementary framework that measures TFP under constant input intensities, while preserving the accounting identity between gross output, value added, and intermediate input expenditures. Second, our framework operates at the sectoral level, distin-

guishing between agriculture, manufacturing, and services, and shows that the implications of time-varying input intensities for TFP measurement differ systematically across sectors due to structurally distinct production technologies. Third, we cover 14 advanced economies, allowing us to document substantial cross-country heterogeneity in both the direction and magnitude of input intensity changes and their consequences for measured productivity.

More broadly, our paper contributes to the literature on structural transformation. A large body of work emphasizes differential productivity growth across sectors as a key driver of structural change (Uy et al., 2013; Swiecki, 2017) and of deindustrialization (Boppart, 2014; Rodrik, 2016; Huneus and Rogerson, 2023; Sposi et al., 2025). Our results suggest that mismeasurement of sectoral productivity arising from changes in intermediate input intensities can affect the quantitative interpretation of sectoral productivity differentials and, consequently, the quantitative implications drawn from structural transformation models with input–output linkages. As shown by Avoumatsodo and Leunga Noukwe (2024) in the case of South Korea, neglecting the evolution of input intensities can lead to substantial overestimation of the effects of productivity shocks on structural transformation.

Finally, we contribute to the literature documenting cross-country differences in intermediate input use. Valentinyi (2021), Sposi (2019), and Boppart et al. (2023) show that input intensities vary systematically across development levels, with poorer countries typically using fewer intermediate inputs than richer ones. By documenting heterogeneous sectoral input intensity dynamics across a broad set of countries and over time, our analysis complements this literature by showing that cross-sectional differences across development levels do not necessarily reflect the within-country evolution of production structures.

Outline. The remainder of the paper is organized as follows. Section 2 introduces the data and presents stylized facts on the evolution of intermediate input intensities across sectors and countries. Section 3 outlines the framework for measuring TFP with time-varying intermediate input intensities. Section 4 presents the quantitative analysis, highlighting the differences between TFP measures under the time-varying and constant input intensity frameworks, with a focus on both long-run trends. Section 5 discusses the results and Section 6 concludes.

2 Heterogeneous Dynamics of Input Intensities

In this section, we document how intermediate input intensities have evolved heterogeneously over time across countries in agriculture, manufacturing, and services.

2.1 Data

To document the stylized facts on the evolution of intermediate input use over time, we draw on the Long-run World Input-Output Database (Woltjer et al., 2021) and the Socio-Economic Accounts (Timmer et al., 2015).⁴ The dataset is particularly well-suited for our purpose as it provides consistent, long-run coverage of sectoral intermediate input expenditures and gross

⁴In the empirical analysis (Section 4), we use alternative data sources, although available for a shorter period (1995–2021), provide richer information on sectoral capital, labor, input use, and labor compensation.

output across a broad set of advanced and emerging economies, thereby allowing us to capture the heterogeneous dynamics of intermediate input intensities over nearly five decades. Based on these sources, we construct a panel dataset covering the period 1965–2014 for 20 countries: Austria, Canada, France, Japan, Korea, Mexico, Portugal, the United Kingdom, India, Italy, the Netherlands, China, Germany, Ireland, Taiwan, Denmark, Finland, the United States, and Sweden. Sectoral intermediate input intensities are computed as the ratio of sectoral intermediate input expenditures to gross output.

To maintain consistency across countries and over time, we adopt the International Standard Industrial Classification, Revision 3 (ISIC Rev. 3) to aggregate industries into three broad sectors: Agriculture: ISIC divisions 1–5 (agriculture, forestry, hunting, and fishing). Manufacturing: ISIC divisions 10–14 (mining and quarrying), 15–37 (manufacturing), 40–41 (utilities), and 45 (construction). Services: ISIC divisions 50–99 (wholesale and retail trade, transport, government, financial, professional, and personal services such as education, health care, and real estate).

2.2 Patterns of Input Intensity Dynamics

Figures 1–3 plot the evolution of intermediate input (II) shares across countries in agriculture, manufacturing, and services over the period 1965–2014. The figures reveal pronounced cross-country heterogeneity in sectoral trajectories. In agriculture, some economies (e.g., Canada, Mexico) exhibit a sharp and persistent rise in input use, while others (e.g., India, the United Kingdom) display stagnation or even decline. Manufacturing patterns are equally diverse: several economies show steady increases (e.g., China, Germany), whereas Asian newly industrialized economies follow U-shaped paths (e.g., Japan, Taiwan, Korea), with shares falling in earlier decades before rising again more recently. Other countries, such as the United Kingdom and Ireland, experience a sustained decline in manufacturing input shares. In services, the picture is more evenly split, with some economies recording gradual increases (e.g., United States, Canada) and others showing flat or declining trajectories (e.g., Brazil, Sweden, France).

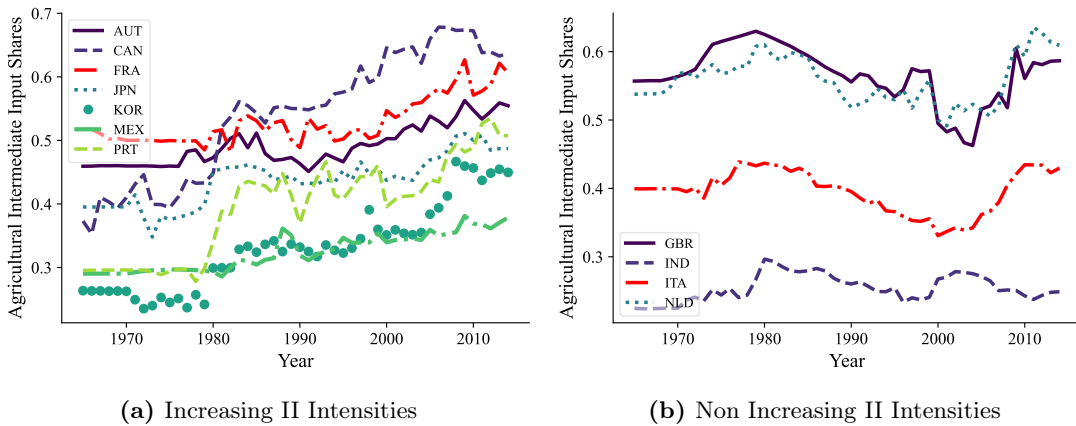


Figure 1: Dynamics of Intermediate Input Intensities in Agriculture Across Countries

Notes: This figure illustrates the evolution of intermediate input shares in agricultural gross output from 1965 to 2014 across 11 countries with significant variation in agricultural intermediate input intensities. Panel (a) shows 7 countries with increasing intermediate input shares over time, and Panel (b) shows 4 countries with non increasing intermediate input shares over time.

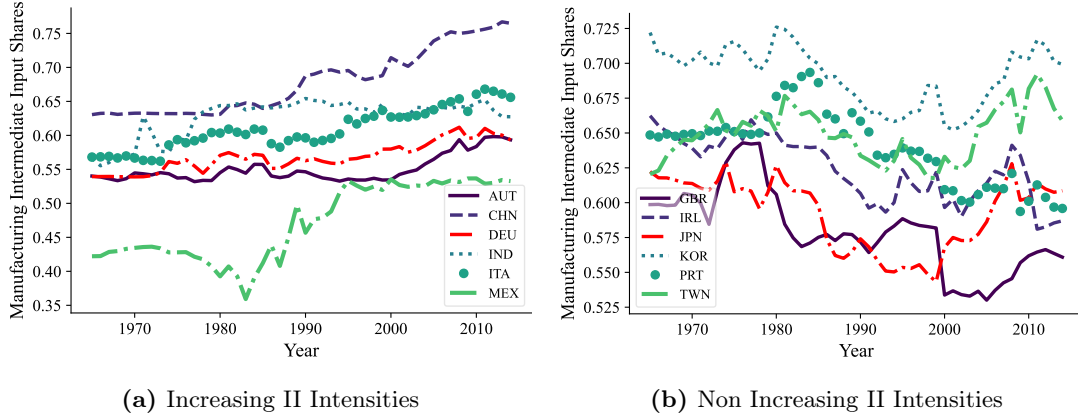


Figure 2: Dynamics of Intermediate Input Intensities in Manufacturing Across Countries

Notes: This figure illustrates the evolution of intermediate input shares in manufacturing gross output from 1965 to 2014 across 12 countries with significant variation in manufacturing intermediate input intensities. Panel (a) shows 6 countries with increasing intermediate input shares over time, and Panel (b) shows 6 countries with U-shaped or decreasing intermediate input intensities over time.

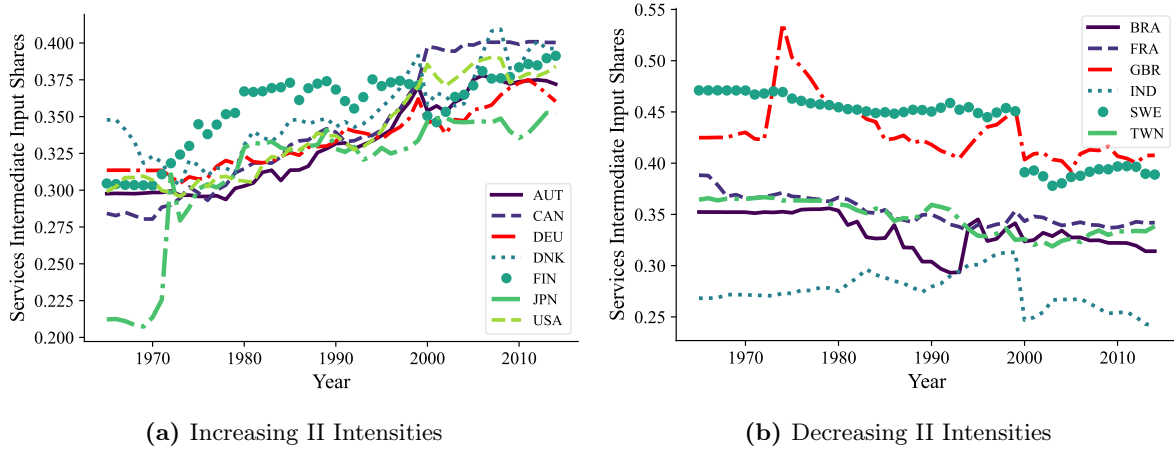


Figure 3: Dynamics of Intermediate Input Intensities in Services Across Countries

Notes: This figure illustrates the evolution of intermediate input shares in services gross output from 1965 to 2014 across 13 countries with significant variation in services intermediate input intensities. Panel (a) shows 7 countries with increasing intermediate input intensities over time, and Panel (b) shows 6 countries with decreasing intermediate input intensities over time.

At first glance, these patterns appear to contradict the view, emphasized in [Sposi \(2019\)](#) and [Boppart et al. \(2023\)](#), that richer economies consistently employ more intermediate inputs. While the cross-sectional ranking of countries broadly supports that finding, the time-series evidence reveals important within-country reversals: high-income economies may experience sustained declines in input intensities, while some middle-income economies undergo rapid increases, and vice versa. If one were to extrapolate from the cross-sectional to the time-series dimension, one would expect input intensities to rise steadily as countries develop or as time progresses, which is not the case. Taken together, the evidence suggests that input intensification is not merely a by-product of development stage but instead reflects deeper, country-specific dynamics.

These empirical findings resonate with recent work emphasizing the centrality of time-varying input-output linkages in shaping structural transformation. [Avoumatsodo and Leunga Noukwe \(2024\)](#) documents substantial evolution in sectoral intermediate input shares in South Korea,

highlighting the importance of accounting for time-varying input-output linkages when modeling structural transformation. Similarly, [Gaggl et al. \(2023\)](#) analyzes structural change in production networks in the United States, revealing a declining fraction of production by goods sectors and a rising fraction by services sectors, and emphasizing the role of evolving input-output networks in these trends. Our study extends these contributions by documenting the dynamic and heterogeneous nature of intermediate input intensities across countries over more than five decades, providing a broader empirical foundation for understanding the non-generalizability of cross-section results on intermediate input use.

These patterns have two implications for productivity measurement. First, because intermediate input and value-added shares jointly determine gross output, shifts in the former directly reshape the decomposition of sectoral gross output growth. Second, assuming constant input shares, as is common in multisector growth models, misses the heterogeneous and the dynamic trajectories documented in the data. Such an assumption can lead to systematic mismeasurement: in some country-sector cases, rising input use may inflate measured productivity growth, while in others, declining shares may understate it.

In sum, the evidence demonstrates that the dynamics of intermediate input intensities are neither monotonic nor universal. This raises a central question: how do these evolving input shares affect the measurement of TFP? As highlighted by [Lipsey and Carlaw \(2004\)](#), TFP measures capture more than pure technological change and can be influenced by shifts in input composition and efficiency. To address this, we next develop a model that explicitly incorporates time-varying input shares, allowing us to quantify how their dynamics shape measured productivity across countries and sectors.

3 Theoretical Framework

This section develops the accounting framework used in the quantitative analysis. We proceed in four steps. We first set up the production environment, describing how representative firms in each sector combine value added and intermediate inputs to produce gross output. We then derive a decomposition of gross output TFP growth into three components: a value-added block, an intermediate-input block, and a term that captures the shift in the relative weight between the two when input intensity changes over time. We next examine what happens to this decomposition when input intensity is held fixed at a constant level, making explicit the source and form of the resulting measurement gap. Finally, we aggregate sectoral TFP growth rates to the economy-wide level using time-varying Domar weights. The decompositions that follow are exact arithmetic identities and hold regardless of what drives input intensity dynamics in the data.

3.1 Production Technology

Setup. Consider sector $n \in \{a, m, s\}$, where a , m , and s denote agriculture, manufacturing, and services. Each sector produces gross output $Y_n(t)$ by combining value added $V_n(t)$ and an

intermediate input bundle $E_n(t)$. The production technology takes the Cobb–Douglas form

$$Y_n(t) = V_n(t)^{1-\lambda_n(t)} E_n(t)^{\lambda_n(t)}, \quad (3.1)$$

where $\lambda_n(t) \in (0, 1)$ denotes the intermediate input intensity, i.e., the share of intermediate inputs in gross output. Value added is produced using capital and labor under a sector-specific value-added productivity level $A_{V,n}(t)$:

$$V_n(t) = A_{V,n}(t) K_n(t)^{\alpha_n} L_n(t)^{1-\alpha_n}, \quad (3.2)$$

where $\alpha_n \in (0, 1)$ denotes the capital share in value added. Substituting Equation (3.2) into Equation (3.1) yields

$$Y_n(t) = (A_{V,n}(t) K_n(t)^{\alpha_n} L_n(t)^{1-\alpha_n})^{1-\lambda_n(t)} E_n(t)^{\lambda_n(t)}. \quad (3.3)$$

The capital share α_n is sector-specific but held constant over time, consistent with standard growth accounting practice. The intermediate input intensity $\lambda_n(t)$ is the central parameter of interest: it is allowed to vary exogenously over time. As documented in Section 2, $\lambda_n(t)$ exhibits substantial variation across countries and sectors.

The production technology in Equation (3.1) assumes a Cobb–Douglas aggregator over value added and intermediate inputs, which imposes a unit elasticity of substitution between the two. This specification has two important implications for the decomposition that follows. First, the expenditure shares of value added and intermediates in gross output are equal to their respective exponents, $(1 - \lambda_n(t))$ and $\lambda_n(t)$, which are directly observable from the data and vary over time in our framework. Second, the Cobb–Douglas production structure yields a tractable and exact decomposition of gross output TFP growth into the three components derived in Section 3.2, with no approximation error. These properties make the Cobb–Douglas specification particularly well suited to the growth accounting exercise we pursue. The specification is also standard in the structural transformation literature ([Herrendorf et al., 2014](#); [Valentinyi, 2021](#); [Sposi, 2019](#); [Gaggl et al., 2023](#)), where it is adopted both for its tractability and for its consistency with balanced growth along a steady-state path.⁵

We treat the intermediate input intensity as an observed, time-varying parameter, not as a structural constant. In practice, $\lambda_n(t)$ reflects the cumulative outcome of firm-level decisions shaped by relative prices, outsourcing opportunities, and technological change. We do not model these determinants. Instead, we take the observed sequence $\{\lambda_n(t)\}$ directly from the data and ask a pure accounting question: how does allowing $\lambda_n(t)$ to vary, rather than fixing it at a constant baseline, affect the measurement of sectoral TFP? The decompositions that follow are exact arithmetic identities and hold regardless of what drives $\lambda_n(t)$ in the data. This approach is consistent with the long tradition of growth accounting as a measurement discipline ([Hulten,](#)

⁵That said, the Cobb–Douglas form is a restriction. If the true elasticity of substitution between value added and intermediate inputs differs from one, the weights in the TFP decomposition would depend not only on observed expenditure shares but also on relative prices and the curvature of the production function, and the exact form of the gap between time-varying and constant-share measures would potentially differ (see, for example, [Aiyar and Dalgaard, 2009](#)).

1978; Corrado et al., 2019).

Intermediate Inputs. Following Gaggi et al. (2023), the intermediate bundle $E_n(t)$ is an aggregate of goods sourced from all sectors, combined via a CES aggregator:

$$E_n(t) = A_{E,n}(t) \left(\sum_{k \in \{a,m,s\}} \omega_{E,nk} E_{nk}(t)^{(\varepsilon_n-1)/\varepsilon_n} \right)^{\varepsilon_n/(\varepsilon_n-1)}, \quad (3.4)$$

where $\varepsilon_n > 0$ is the elasticity of substitution between intermediate inputs, $\omega_{E,nk} \geq 0$ is the weight of inputs sourced from sector k (with $\sum_k \omega_{E,nk} = 1$), and $A_{E,n}(t)$ is an efficiency shifter for the intermediate bundle.⁶ $E_{nk}(t)$ denotes the quantity of the final good from sector k used in the production of the sector n good. The intermediate bundle corresponding price index is

$$P_{E,n}(t) = \frac{1}{A_{E,n}(t)} \left(\sum_{k \in \{a,m,s\}} \omega_{E,nk}^{\varepsilon_n} P_k(t)^{1-\varepsilon_n} \right)^{1/(1-\varepsilon_n)}, \quad (3.5)$$

where $P_k(t)$ is the price of the final good produced by sector k .

Equilibrium Conditions. Under perfect competition, profit maximization by the producer of the final good in sector n implies that each factor is paid its marginal product:

$$R_n(t) = \alpha_n (1 - \lambda_n(t)) P_n(t) \frac{Y_n(t)}{K_n(t)}, \quad (3.6)$$

$$W_n(t) = (1 - \alpha_n) (1 - \lambda_n(t)) P_n(t) \frac{Y_n(t)}{L_n(t)}, \quad (3.7)$$

$$P_{E,n}(t) = \lambda_n(t) P_n(t) \frac{Y_n(t)}{E_n(t)}, \quad (3.8)$$

$$P_{V,n}(t) = (1 - \lambda_n(t)) P_n(t) \frac{Y_n(t)}{V_n(t)}. \quad (3.9)$$

where $R_n(t)$ and $W_n(t)$ are the rental rate of capital and the wage, $P_{E,n}(t)$ is the intermediate bundle input price index, and $P_{V,n}(t)$ is the value-added deflator. These conditions link factor payments to output shares and will be used throughout the derivation below.

Although this analysis is conducted within a partial-equilibrium accounting framework, the following market-clearing conditions describe the allocation of resources:

$$\sum_{n \in \{a,m,s\}} L_n(t) = L(t), \quad \sum_{n \in \{a,m,s\}} K_n(t) = K(t), \quad Y_n(t) = \sum_{k \in \{a,m,s\}} E_{kn}(t) + C_n(t),$$

⁶Intermediate input efficiency $A_{E,n}(t)$ should be understood as a reduced-form representation of how effectively a sector integrates intermediate inputs into production. It may reflect differences in procurement technology, supply-chain coordination, input quality, and the degree of specialization in production networks. Such efficiency is plausibly heterogeneous across sectors: services, for instance, tend to rely more on relational and localized input structures, while manufacturing often operates through more standardized and globally integrated supply chains, and agriculture depends more heavily on seasonally and geographically constrained inputs.

where $C_n(t)$ is household consumption of sector- n goods.

3.2 TFP Growth Decomposition

Taking logarithms of Equation (3.1) and differentiating with respect to time yields:

$$\frac{\dot{Y}_n(t)}{Y_n(t)} = (1 - \lambda_n(t)) \frac{\dot{V}_n(t)}{V_n(t)} + \lambda_n(t) \frac{\dot{E}_n(t)}{E_n(t)} + \dot{\lambda}_n(t) \log\left(\frac{E_n(t)}{V_n(t)}\right), \quad (3.10)$$

where $\dot{X}(t)/X(t)$ denotes the growth rate of a variable X and $\dot{\lambda}_n(t)$ captures the dynamics of intermediate input intensity. This decomposition has three terms. The first two are standard: gross output grows through growth in value added, weighted by $(1 - \lambda_n)$, and through growth in intermediates, weighted by λ_n . The third term arises when the intermediate input intensity varies over time. When input intensity changes ($\dot{\lambda}_n \neq 0$), the weights on value added and intermediates shift over time, and this shift independently contributes to output growth. Fixing λ_n at a constant makes this term vanish by assumption, and that is precisely a source of mismeasurement.

Value-Added TFP. Differentiating Equation (3.2) gives the growth rate of value added:

$$\frac{\dot{V}_n(t)}{V_n(t)} = \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} + \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} + (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)}. \quad (3.11)$$

Value-added TFP growth, $\dot{A}_{V,n}/A_{V,n}$, is the familiar Solow residual: value-added growth less the factor-share-weighted growth rates of capital and labor.

Intermediate-Input Efficiency. Growth in the intermediate bundle has an analogous decomposition. Letting $s_{E,nk}(t) \equiv P_k(t)E_{nk}(t)/P_{E,n}(t)E_n(t)$ denote the cost share of inputs sourced from sector k , the CES aggregator properties imply:⁷

$$\frac{\dot{E}_n(t)}{E_n(t)} = \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} + \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \frac{\dot{E}_{nk}(t)}{E_{nk}(t)}, \quad (3.12)$$

where $\dot{A}_{E,n}/A_{E,n}$ captures changes in the efficiency with which intermediate inputs sourced from different sectors are aggregated in sector n . The coefficients $s_{E,nk}(t)$ capture the sectoral linkages between sectors in the model.

Sectoral Gross-Output TFP. Substituting Equations (3.11) and (3.12) into Equation (3.10), we obtain the expression for the sectoral gross-output Solow residual. The growth rate of gross-output TFP in sector n , denoted by $A_n(t)$, is defined as the growth of gross output net of the contribution of primary and intermediate inputs:

$$\frac{\dot{A}_n(t)}{A_n(t)} \equiv \frac{\dot{Y}_n(t)}{Y_n(t)} - (1 - \lambda_n(t)) \left[\alpha_n \frac{\dot{K}_n(t)}{K_n(t)} + (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)} \right] - \lambda_n(t) \sum_k s_{E,nk}(t) \frac{\dot{E}_{nk}(t)}{E_{nk}(t)}.$$

⁷See Appendix B for the proof.

This decomposition makes explicit that gross-output productivity growth is measured as residual output growth after accounting for the contributions of capital, labor, and sector-specific intermediate inputs, weighted by their respective cost shares. A direct computation then yields the central expression:

$$\frac{\dot{A}_n(t)}{A_n(t)} = \underbrace{(1 - \lambda_n(t)) \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)}}_{\text{(i) VA-block TFP}} + \underbrace{\lambda_n(t) \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)}}_{\text{(ii) Intermediate-block TFP}} + \underbrace{\dot{\lambda}_n(t) \log\left(\frac{\lambda_n(t)}{1 - \lambda_n(t)} \frac{P_{V,n}(t)}{P_{E,n}(t)}\right)}_{\text{(iii) Input-intensity shift effect}}. \quad (3.13)$$

The last expression uses the equilibrium conditions (3.8)–(3.9), which together imply $E_n/V_n = \lambda_n/(1 - \lambda_n) \times P_{V,n}/P_{E,n}$. This three-way decomposition of gross-output TFP into a value-added block, an intermediate-input block, and an input-intensity shift term parallels the aggregate decomposition framework in [Balk \(2020\)](#), who derives exact relationships between gross-output and value-added TFP change using Domar factors under general index number assumptions; our framework differs in that we allow the intermediate input weight $\lambda_n(t)$ to vary over time rather than treating it as a structural constant.

Note that $A_n(t)$ does not appear as a primitive in the level production function in Equation (3.1): because $A_{V,n}$ and $A_{E,n}$ are already embedded in V_n and E_n , the model fits the data exactly in levels by construction. Gross-output TFP $A_n(t)$ is therefore defined implicitly through its growth rate in Equation (3.13). In the quantitative analysis, TFP levels are recovered by integrating this growth rate from a common base period, so that the base-period choice anchors the level comparison but does not affect growth rates or the measured gap between the time-varying and constant-intensity scenarios.

3.3 Aggregation via Domar Weights

To measure economy-wide productivity, we aggregate across sectors following [Hulten \(1978\)](#). Let nominal GDP be defined as the sum of nominal sectoral value added:

$$P_V(t) V(t) \equiv \sum_{n \in \{a, m, s\}} P_{V,n}(t) V_n(t), \quad (3.14)$$

where $V(t)$ denotes real GDP and $P_V(t)$ is the GDP deflator at time t .

The Domar weight of sector n is defined as the share of its nominal gross output in nominal GDP:

$$d_n(t) \equiv \frac{P_n(t) Y_n(t)}{P_V(t) V(t)}. \quad (3.15)$$

Equivalently, using the decomposition of value added into gross output net of intermediate inputs, we can write

$$d_n(t) = \frac{P_n(t) Y_n(t)}{\sum_{k \in \{a, m, s\}} (1 - \lambda_k(t)) P_k(t) Y_k(t)}, \quad (3.16)$$

so that the Domar weights evolve over time both with changes in sectoral gross output and with shifts in intermediate input shares, reflecting changes in the relative importance of sectors in the economy.⁸

⁸[Diewert \(2015\)](#) provides complementary decompositions of economy-wide TFP growth into sectoral effects,

Aggregate Gross-Output TFP. As shown by [Hulten \(1978\)](#), [Ten Raa and Shestalova \(2011\)](#), [Baqae and Farhi \(2019\)](#), and consistency with [Esfahani et al. \(2024\)](#), aggregate TFP growth is the Domar-weighted sum of sectoral TFP growth rates:

$$\frac{\dot{A}(t)}{A(t)} = \sum_{n \in \{a, m, s\}} d_n(t) \frac{\dot{A}_n(t)}{A_n(t)}. \quad (3.17)$$

When $\lambda_n(t)$ increases, the Domar weight of sector n declines, *ceteris paribus*, reflecting the lower share of value added in gross output. Variations in input intensity therefore shape aggregate productivity dynamics through two complementary channels: (i) changes in Domar weights, and (ii) changes in measured sectoral TFP growth arising from evolving production structures.

Aggregate Value-Added TFP. Complementing the gross-output measure, aggregate value-added TFP growth is the nominal-value-added-share-weighted average of sectoral value-added TFP:

$$\frac{\dot{A}_V(t)}{A_V(t)} \equiv \sum_{n \in \{a, m, s\}} s_{V,n}(t) \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)}, \quad (3.18)$$

where $s_{V,n}(t) \equiv P_{V,n}(t)V_n(t)/P_V(t)V(t)$ is the nominal value-added share of sector n . We also derive consistent aggregate factor-input growth indices⁹. To do so, let us define the aggregate capital share

$$\alpha(t) \equiv \sum_n s_{V,n}(t) \alpha_n.$$

The following aggregate decomposition then holds exactly:

$$\frac{\dot{V}(t)}{V(t)} = \frac{\dot{A}_V(t)}{A_V(t)} + \alpha(t) \frac{\dot{K}(t)}{K(t)} + (1 - \alpha(t)) \frac{\dot{L}(t)}{L(t)}, \quad (3.19)$$

where aggregate capital and labor growth rates are given by

$$\frac{\dot{K}(t)}{K(t)} \equiv \frac{1}{\alpha(t)} \sum_n s_{V,n}(t) \alpha_n \frac{\dot{K}_n(t)}{K_n(t)}, \quad \text{and} \quad \frac{\dot{L}(t)}{L(t)} \equiv \frac{1}{1 - \alpha(t)} \sum_{n \in \{a, m, s\}} s_{V,n}(t) (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)}.$$

Equation (3.19) shows that aggregate value-added growth admits exactly the standard Solow decomposition. In this sense, the aggregate economy “recovers” the canonical structure once inputs are properly aggregated. Equations (3.11)–(3.19) then form a complete, internally consistent accounting system. They make explicit, at both the sectoral and aggregate level, how the treatment of intermediate input intensities shapes measured productivity. Section 4 quantifies these effects in the data.

including explicit terms for changes in sectoral input shares and real input prices. Their framework confirms that the weights in the Domar aggregation are themselves affected by changes in input shares, consistent with our Equation (3.16).

⁹Equation (3.18) Divisia growth in aggregate value-added TFP is value-added-weighted growth in sectoral real value-added TFP.

4 Quantitative Analysis

This section implements the growth accounting framework developed in Section 3. The goal is to quantify the measurement gap that arises from treating intermediate input intensities as constant rather than time-varying. We proceed in two steps. We first describe how the model is mapped to the data, making explicit how each variable and parameter is constructed under the two scenarios. We then describe the data sources and the aggregation procedure.

4.1 Mapping Model to Data

The accounting exercise compares productivity measures constructed under two assumptions about the intermediate input intensity $\lambda_n(t)$:

- *Time-varying scenario*: $\lambda_n(t)$ takes its observed value in each period, computed from the data as described below.
- *Constant-intensity scenario*: $\lambda_n(t)$ is held fixed at its sample mean $\bar{\lambda}_n$ throughout the period.

Capital, labor, and gross output are held at their observed values in both scenarios. Any difference in measured productivity between the two scenarios is therefore attributable solely to the treatment of $\lambda_n(t)$, cleanly isolating the contribution of time-varying input intensities to measured TFP.

Let $X(t)$ be a generic model variable and $\widetilde{X}(t)$ its corresponding observed data counterpart. The sectoral capital income share α_n is calibrated as one minus the average labor compensation share in value added:

$$\alpha_n = 1 - \frac{1}{T} \sum_{t=1}^T \frac{\widetilde{W}_n(t) \widetilde{L}_n(t)}{\widetilde{P}_{V,n}(t) \widetilde{V}_n(t)}, \quad (4.1)$$

where $\widetilde{W}_n(t) \widetilde{L}_n(t)$ is sectoral observed labor compensation, and $\widetilde{P}_{V,n}(t) \widetilde{V}_n(t)$ is nominal sectoral value added. Averaging over time yields a single, time-invariant parameter per sector and country.

Measuring Intermediate-Input Intensities. In the time-varying scenario, intermediate input intensity is constructed period by period as the ratio of observed nominal intermediate input expenditures to nominal gross output:

$$\lambda_n(t) = \frac{\widetilde{P}_{E,n}(t) \widetilde{E}_n(t)}{\widetilde{P}_n(t) \widetilde{Y}_n(t)}, \quad \forall t = 1, \dots, T. \quad (4.2)$$

This ratio is directly observable and corresponds exactly to the equilibrium condition in Equation (3.8).

In the constant-intensity scenario, $\lambda_n(t)$ is replaced by its sample mean:

$$\bar{\lambda}_n = \frac{1}{T} \sum_{t=1}^T \frac{\widetilde{P}_{E,n}(t) \widetilde{E}_n(t)}{\widetilde{P}_n(t) \widetilde{Y}_n(t)}. \quad (4.3)$$

Using $\bar{\lambda}_n$ introduces an internal consistency constraint. Because $\lambda_n(t)$ is defined as the ratio of nominal intermediate expenditures to nominal gross output, fixing $\lambda_n(t) = \bar{\lambda}_n$ while keeping gross output at its observed value implies that intermediate expenditures must equal $\bar{\lambda}_n \cdot \widetilde{P}_n(t) \widetilde{Y}_n(t)$. This quantity differs from what is observed in the data whenever $\bar{\lambda}_n \neq \lambda_n(t)$.

National accounts systems are constructed so that nominal gross output, nominal value added, and nominal intermediate expenditures satisfy the identity $P_n Y_n = P_{V,n} V_n + P_{E,n} E_n$ at every point in time. When intermediate input intensity $\lambda_n(t)$ varies but is artificially held fixed, this identity can no longer hold simultaneously for all three observed series. Enforcing it requires rescaling one of the three components, and since gross output and intermediate expenditures are taken directly from the data, the adjustment falls on value added, which is exactly the channel through which constant-intensity approaches mismeasure productivity. Importantly, this gap is not an artifact; it captures how much of the total measurement gap is attributable to the internal consistency adjustment and how much to the input-intensity shift effect. Therefore, real value added is then determined residually as

$$V_n(t) = \frac{(1 - \bar{\lambda}_n) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_{V,n}(t)} \quad (4.4)$$

which deviates from its empirical counterpart whenever $\bar{\lambda}_n \neq \lambda_n(t)$.

National accounts systems are constructed so that nominal gross output, nominal value added, and nominal intermediate expenditures satisfy the identity $P_n Y_n = P_{V,n} V_n + P_{E,n} E_n$ at every point in time. When intermediate input intensity $\lambda_n(t)$ varies but is artificially held fixed, this identity can no longer hold simultaneously for all three observed series. Enforcing it requires rescaling one of the three components, and since gross output and intermediate expenditures are taken directly from the data, the adjustment falls on value added, which is exactly the channel through which constant-intensity approaches mismeasure productivity. Importantly, this gap is not an artifact; it captures how much of the total measurement gap is attributable to the internal consistency adjustment and how much to the input-intensity shift effect.

For comparability with the structural change literature, which typically assumes constant shares without adjusting the data (see, for example, [Uy et al., 2013](#); [Lewis et al., 2022](#)), we also report productivity measures under fixed $\bar{\lambda}_n$ without any data adjustment. In this case, as shown Appendix B, the productivity growth difference between the two approaches arises mainly from the input-share shift effect, $\dot{\lambda}_n(t) \phi_n(t)$, where

$$\phi_n(t) \equiv \log \left(\frac{\lambda_n(t)}{1 - \lambda_n(t)} \frac{P_{V,n}(t)}{P_{E,n}(t)} \right). \quad (4.5)$$

Value-Added TFP. In discrete time, Equation (3.11) yields value-added TFP growth:

$$\Delta \log A_{V,n}(t) = \Delta \log V_n(t) - \alpha_n \Delta \log \widetilde{K}_n(t) - (1 - \alpha_n) \Delta \log \widetilde{L}_n(t), \quad (4.6)$$

where $\Delta \log X(t) \equiv \log X(t) - \log X(t-1)$. Capital $\widetilde{K}_n(t)$ and labor $\widetilde{L}_n(t)$ are taken directly from the data. Under the time-varying scenario, model-implied value added equals its empirical counterpart by construction. Under the constant-intensity scenario with data adjustment, it does

not—and the gap translates directly into a gap in measured productivity. A direct calculation gives:

$$\Delta \log A_{V,n}(t) - \Delta \log \bar{A}_{V,n}(t) = \log(1 - \lambda_n(t)) - \log(1 - \bar{\lambda}_n), \quad (4.7)$$

where $\Delta \log \bar{A}_{V,n}(t)$ is the value-added TFP growth under constant intermediate input intensity. The mismeasurement gap in value-added TFP is fully determined by the deviation of the time-varying $\lambda_n(t)$ from its mean, and grows with the amplitude of the variation in $\lambda_n(t)$.

Aggregate value-added TFP growth is the Törnqvist-weighted average of sectoral growth rates:

$$\Delta \log A_V(t) \equiv \sum_{n \in \{a,m,s\}} \bar{s}_{V,n}(t) \Delta \log A_{V,n}(t), \quad (4.8)$$

where $\bar{s}_{V,n}(t) \equiv [s_{V,n}(t) + s_{V,n}(t-1)]/2$ and

$$s_{V,n}(t) = \frac{(1 - \lambda_n(t)) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\sum_{k \in \{a,m,s\}} (1 - \lambda_k(t)) \widetilde{P}_k(t) \widetilde{Y}_k(t)}. \quad (4.9)$$

The value-added share $s_{V,n}(t)$ is itself affected by the assumption on $\lambda_n(t)$, so the treatment of intermediate input intensities shapes aggregate value-added productivity through two channels simultaneously: sectoral value-added TFP growth rates and the aggregation weights, $s_{V,n}(t)$. For instance, if agriculture's input intensity rises while the other sectors remain constant, the constant-intensity model overstates agriculture's value-added share and consequently overstates its contribution to aggregate productivity.

Intermediate-Input Efficiency. Using Equation (3.12), intermediate-input efficiency growth for sector n is:

$$\Delta \log A_{E,n}(t) = \Delta \log E_n(t) - \sum_{k \in \{a,m,s\}} \bar{s}_{E,nk}(t) \Delta \log E_{nk}(t), \quad (4.10)$$

where real intermediate quantities are recovered from the data as:

$$E_n(t) = \frac{\lambda_n(t) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_{E,n}(t)}, \quad E_{nk}(t) = \frac{\widetilde{s}_{E,nk}(t) \lambda_n(t) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_k(t)}. \quad (4.11)$$

The sectoral input-output coefficients, $\widetilde{s}_{E,nk}(t)$, are taken directly from the data and held identical across both scenarios, so that differences in TFPs arise solely from the treatment of $\lambda_n(t)$.

Gross-Output TFP. In discrete time, Equation (3.13) becomes:

$$\Delta \log A_n(t) = (1 - \bar{\lambda}_n(t)) \Delta \log A_{V,n}(t) + \bar{\lambda}_n(t) \Delta \log A_{E,n}(t) + \Delta \lambda_n(t) \bar{\phi}_n(t), \quad (4.12)$$

where $\bar{\lambda}_n(t) \equiv [\lambda_n(t) + \lambda_n(t-1)]/2$, $\Delta \lambda_n(t) \equiv \lambda_n(t) - \lambda_n(t-1)$, and $\bar{\phi}_n(t)$ is the two-period average of $\phi_n(t)$. Under the constant-intensity scenario without data adjustment, the input-intensity shift term $\Delta \lambda_n(t) \bar{\phi}_n(t)$ vanishes and the weights revert to constants $\bar{\lambda}_n$ and $1 - \bar{\lambda}_n$.

Aggregate gross-output TFP growth is obtained by Domar aggregation:

$$\Delta \log A(t) = \sum_{n \in \{a, m, s\}} \bar{d}_n(t) \Delta \log A_n(t), \quad (4.13)$$

where

$$d_n(t) = \frac{\widetilde{P}_n(t) \widetilde{Y}_n(t)}{\sum_{k \in \{a, m, s\}} (1 - \lambda_k(t)) \widetilde{P}_k(t) \widetilde{Y}_k(t)}, \quad (4.14)$$

and $\bar{d}_n(t)$ is the two-period average of

4.2 Data

The WIOD long-run data used in Section 2 span nearly five decades and exhibit substantial variation in intermediate input intensities, making them well suited to illustrate how these changes relate to productivity dynamics. However, the WIOD database does not provide the sectoral capital and labor series required to construct TFP measures. We therefore rely on the EUKLEMS & INTANProd 2025 Release, which covers a shorter period (1995–2021) but provides comprehensive and harmonized data on capital services, labor compensation, and factor inputs, for 38 industries, classified according to NACE Rev. 2. The panel covers 14 countries: Austria, Belgium, Czech Republic, Germany, Denmark, Finland, France, Greece, Hungary, Latvia, Luxembourg, Netherlands, Japan, and the United States. Countries with substantial missing observations in capital or labor series were excluded.

To construct the sectoral linkage coefficients, $s_{E,nk}(t)$, we complement the EUKLEMS data with information from the OECD Inter-Country Input-Output (ICIO) Tables (OECD, 2023). The ICIO database provides internationally harmonized input-output tables describing flows of intermediate inputs across industries and countries. Using these tables, we compute the intermediate input coefficients $s_{E,nk}(t)$, defined as the share of inputs from sector k used in the production of sector n . Formally, these coefficients are calculated as the ratio of intermediate purchases from sector k by sector n to the total intermediate inputs used by sector n , over the period 1995–2020.

Sector Aggregation. We aggregate the 38 industries into three broad sectors.¹⁰ This three-sector aggregation serves two purposes. First, it ensures consistency with the stylized-facts section and with the structural transformation literature, which standardly uses this classification for cross-country comparability (Herrendorf et al., 2014; Valentinyi, 2021). Second, it keeps the analysis tractable across 14 countries and 25 years while retaining the key sectoral contrasts between Agriculture, Manufacturing, and services.

We acknowledge that aggregation into three sectors conceals heterogeneity within each broad category: software services and retail trade, for instance, have markedly different intermediate input intensities, yet both fall under Services. This is a known limitation of the approach,

¹⁰Agriculture covers NACE division A (agriculture, forestry, and fishing). Manufacturing covers B (mining and quarrying), C (manufacturing), D (electricity, gas, steam, and air conditioning supply), E (water supply, sewerage, waste management, and remediation), and F (construction). Services cover all remaining industries, G through U.

shared with most of the structural transformation literature. To mitigate it, the aggregation procedure described below preserves within-sector heterogeneity in the construction of sectoral growth rates, by weighting each industry by its nominal value share before aggregating.

Aggregation Procedure. Employment and all nominal variables are aggregated to the sector level by summation. For real variables, simple summation is exact only in the reference year, when nominal and real values coincide. For all other years, sectoral real quantities are constructed via Törnqvist indexes applied to the reference-year benchmarks. Specifically, for any variable x , let $x_n^j(t)$ denote the real quantity in industry $j \in \{1, \dots, N_n\}$ within sector n at time t , and $X_n(t)$ the sectoral aggregate real value. The growth rate of $X_n(t)$ is:

$$\Delta \log X_n(t) = \sum_{j=1}^{N_n} \bar{s}_n^j(t) (\log x_n^j(t) - \log x_n^j(t-1)), \quad (4.15)$$

where

$$s_n^j(t) \equiv \frac{p_{x,n}^j(t) x_n^j(t)}{\sum_{i=1}^{N_n} p_{x,n}^i(t) x_n^i(t)}, \quad \bar{s}_n^j(t) \equiv \frac{1}{2} (s_n^j(t) + s_n^j(t-1)),$$

with $p_{x,n}^j(t)$ the price index for industry j in sector n . Sectoral level series are then recovered by applying these growth rates to the reference-year benchmark values.

4.3 Results

Figures 4 and 5 plot measured sectoral and aggregate TFP for each country under two scenarios: constant input intensity (dashed blue line) and time-varying input intensity (solid black line), with an additional series showing the constant-intensity measure without data adjustment (dashed-marked red line). Across all countries and sectors, the treatment of intermediate input intensities systematically shapes measured productivity. Table 1 quantifies these discrepancies as of 2020. Column (1) reports the percentage change in $\lambda_n(t)$ between 1995 and 2020. Column (2) reports the TFP gap (defined as time-varying TFP minus constant-intensity TFP) when the data are adjusted to maintain internal consistency (as described in Section 4.1). Column (3) reports the same gap when the data are left unadjusted. A positive gap indicates that TFP is higher under the time-varying scenario; a negative gap indicates that TFP is higher under the constant-share scenario, so the latter overstates productivity.

Columns (2) and (3) answer distinct and complementary questions. Column (2) addresses the question: by how much does measured TFP change when intermediate input intensities are allowed to vary over time, relative to a benchmark that holds them fixed while maintaining full consistency with the national accounts identity between nominal gross output, nominal value added, and nominal intermediate expenditures? This is the internally consistent comparison. Column (2) therefore represents the methodologically consistent benchmark against which the time-varying measure should be evaluated.

Column (3) addresses a different question: how sensitive are the TFP estimates produced by the standard approach in the structural transformation literature to the treatment of intermediate input intensity? In that scenario, input shares are typically held fixed without adjusting

the underlying data series for consistency. Column (3) replicates this approach exactly, using the same observed data series under both the time-varying and constant-intensity assumptions, so that any difference between the two measures isolates the contribution of the input-intensity shift effect alone, as derived in Equation (B.13) of Appendix B. Comparing columns (2) and (3) therefore, reveals how much of the total measurement gap is attributable to the internal consistency adjustment and how much to the input-intensity shift effect, providing a more complete picture of the sources of gap between the two approaches than either column alone could deliver.

Table 1: TFP Differences Under Time-Varying and Constant Input Shares, 2020

Country	Agriculture			Manufacturing			Services			Aggregate	
	% $\Delta\lambda$	% Δ TFP		% $\Delta\lambda$	% Δ TFP		% $\Delta\lambda$	% Δ TFP		% Δ TFP	
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)	(2)	(3)
Austria	14.83	-3.99	2.13	14.58	-4.05	4.36	12.76	-6.43	-2.37	-10.69	-3.27
Belgium	16.51	0.53	11.72	5.33	0.74	4.48	4.08	-0.89	-0.44	-0.11	3.69
Czech Republic	10.89	-5.28	0.89	4.18	-1.35	2.44	-5.34	2.65	0.34	1.27	6.10
Germany	8.38	-4.91	1.47	6.54	-1.78	1.70	17.34	-9.61	-3.26	-12.55	-9.65
Denmark	17.41	-2.28	6.91	1.22	0.13	1.09	15.93	-9.29	-3.13	-11.52	-10.04
Finland	8.07	-2.26	-0.35	3.47	0.17	1.59	8.80	-4.57	-1.39	-5.69	-3.31
France	4.16	-0.36	1.45	5.85	-0.69	2.93	11.33	-6.51	-2.23	-8.65	-6.05
Greece	13.32	-9.18	-3.88	1.00	0.02	0.05	-9.43	3.90	0.93	5.52	4.44
Hungary	-5.21	2.88	-1.58	5.55	-1.01	2.00	-6.28	3.01	0.91	2.33	5.51
Japan	20.32	-10.34	1.76	3.34	-0.58	1.25	10.85	-5.69	-2.21	-6.51	-4.88
Luxembourg	42.30	-12.42	8.95	14.03	-3.04	7.10	46.62	-12.31	12.66	-26.18	-21.52
Latvia	8.62	-1.11	3.23	12.35	0.08	5.61	-13.16	6.38	0.97	8.18	13.60
Netherlands	16.37	-3.77	2.78	9.93	-1.08	4.53	12.39	-5.91	-1.40	-8.87	-4.69
United States	-4.21	0.71	-1.90	-8.14	3.20	-1.40	3.79	-2.18	-0.82	-1.02	-3.19

Notes: This table reports the TFP gap in 2020 between two approaches: (i) time-varying intermediate input shares and (ii) constant intermediate input shares fixed at their sample averages. Column (1) reports the percentage change in the sectoral intermediate input intensity λ_n between 1995 and 2020. Column (2) shows the gap when the underlying data are adjusted for internal consistency (Section 4.1), and column (3) shows the gap without data adjustment. A positive gap indicates that TFP is higher under the time-varying scenario; a negative gap indicates that TFP is higher under the constant-share scenario, so the latter overstates productivity.

A clear and systematic pattern runs through Table 1. In column (2), the sign of the TFP gap tracks the direction of the λ change in column (1) across all three sectors: when input intensity rises, the constant-share approach overstates TFP (negative gap); when input intensity falls, it understates TFP (positive gap). This is the direct prediction of Equation (4.7): rising $\lambda_n(t)$ reduces the value-added share below its time-average, so a constant-share model assigns too much of gross output to value added, inflating measured productivity. The size of the overstatement is proportional to the amplitude of the λ change.

The largest overstatements arise where input intensity grew most. Luxembourg has the most extreme case for the period 1995–2020: its agricultural λ rose by 42.30 percentage points and the adjusted TFP gap is -12.42% , meaning that constant-share methods overstate agricultural productivity by 12.42 percentage points by 2020. Japan ($\Delta\lambda = +20.32\%$, gap = -10.34%) and Greece ($\Delta\lambda = +13.32\%$, gap = -9.18%) follow. At the other end, the two countries whose agricultural input intensity declined over 1995–2020—Hungary ($\Delta\lambda = -5.21\%$, gap = $+2.88\%$)

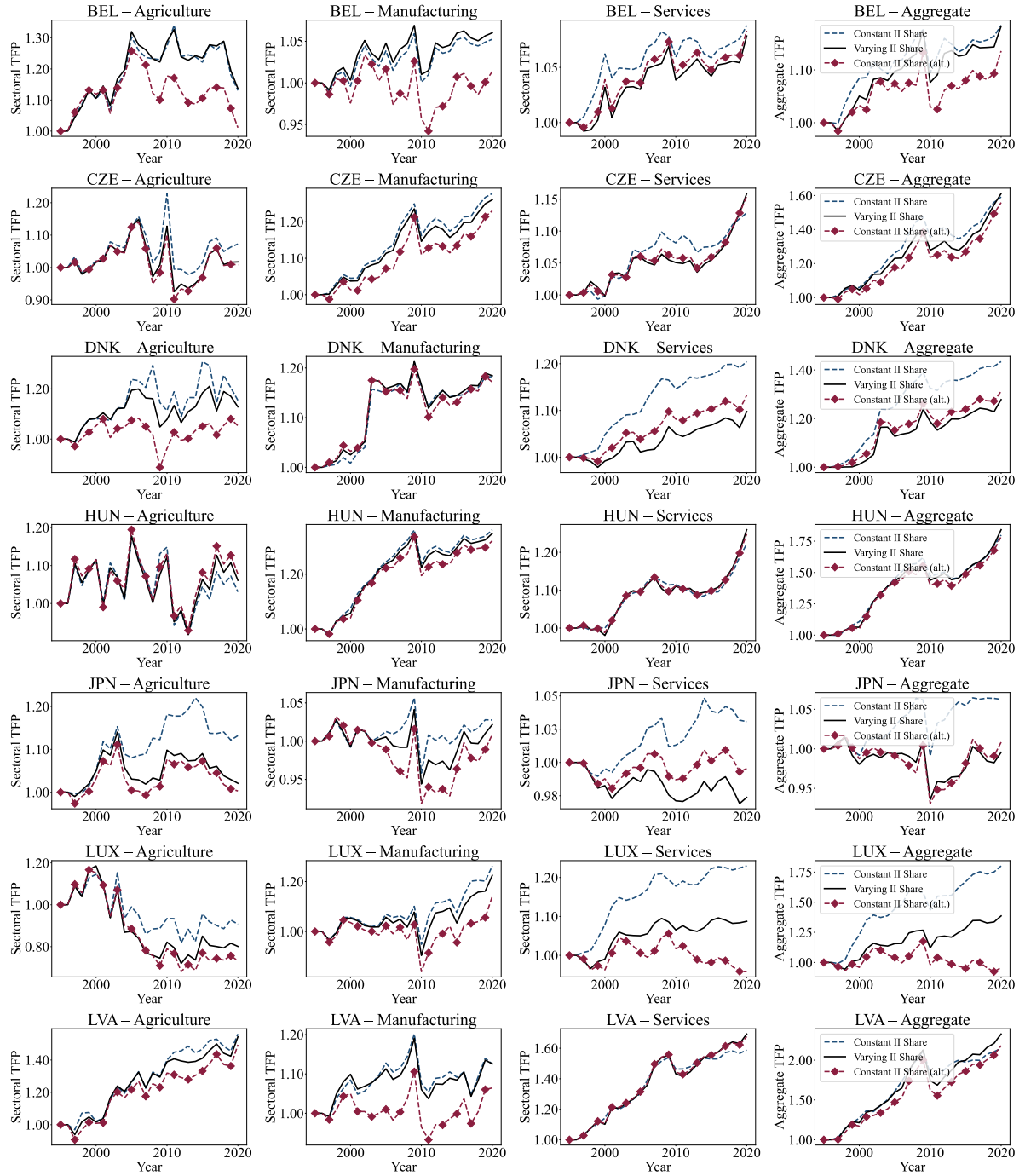


Figure 4: Sectoral and Aggregate TFP Trends – Panel 1

Notes: This figure displays sectoral and aggregate TFP indices (1995=100) over time for Belgium (BEL), Czech Republic (CZE), Denmark (DNK), Hungary (HUN), Japan (JPN), Luxembourg (LUX), and Latvia (LVA). Each row shows agriculture, manufacturing, services, and aggregate TFP from left to right. The solid black line is TFP computed with time-varying intermediate input intensities. The dashed blue line is TFP computed with constant input intensities and adjusted data. The dashed-marked red line is TFP computed with constant input intensities without adjusting the underlying data.

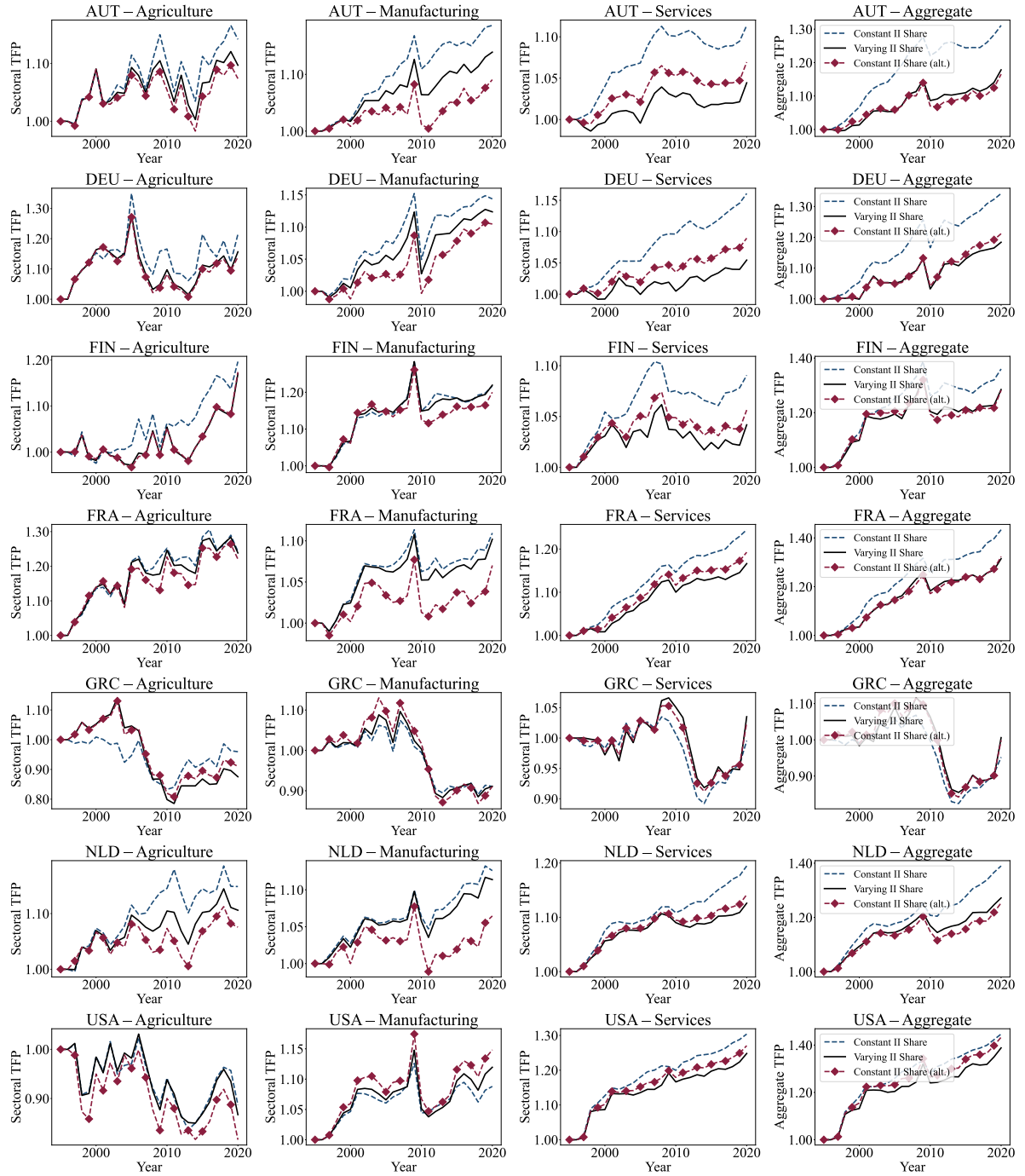


Figure 5: Sectoral and Aggregate TFP Trends – Panel 2

Notes: This figure displays sectoral and aggregate TFP indices (1995=100) over time for Austria (AUT), Germany (DEU), Finland (FIN), France (FRA), Greece (GRC), Netherlands (NLD), and United States (USA). Each row shows agriculture, manufacturing, services, and aggregate TFP from left to right. The solid black line is TFP computed with time-varying intermediate input intensities. The dashed blue line is TFP computed with constant input intensities and adjusted data. The dashed-marked red line is TFP computed with constant input intensities without adjusting the underlying data.

and the United States ($\Delta\lambda = -4.21\%$, gap = $+0.71\%$)—display positive gaps, confirming that falling λ leads to understatement of TFP growth.

Gaps in manufacturing are generally smaller in magnitude, but the same sign pattern holds. Countries with rising manufacturing λ show overstatement: Austria ($\Delta\lambda = +14.58\%$, gap = -4.05%), Luxembourg ($+14.03\%$, -3.04%), Netherlands ($+9.93\%$, -1.08%), and Germany ($+6.54\%$, -1.78%). Belgium, Denmark, Finland, and Latvia show near-zero adjusted gaps ($+0.74\%$, $+0.13\%$, $+0.17\%$, $+0.08\%$), consistent with modest changes in their manufacturing λ over the period 1995–2020.

Services displays the largest gaps in absolute value for many countries. Germany ($\Delta\lambda = +17.34\%$, gap = -9.61%), Denmark ($+15.93\%$, -9.29%), and Luxembourg ($+46.62\%$, -12.31%) stand out as particularly large overstatements.

Countries where service-sector λ fell all exhibit positive gaps: Czech Republic ($\Delta\lambda = -5.34\%$, gap = $+2.65\%$), Hungary (-6.28% , $+3.01\%$), Greece (-9.43% , $+3.90\%$), and Latvia (-13.16% , $+6.38\%$).

The no-adjustment case (column 3) isolates the input-intensity shift effect. For services, it shows a similar but attenuated pattern compared with column (2). For agriculture and manufacturing, the sign reverses because, unlike in services where the input-intensity shift effect reinforces the value-added TFP biases, it partially offsets them in goods-producing sectors.¹¹

Aggregating through Domar weights, the cumulative overstatement is largest where multiple sectors simultaneously exhibit rising input intensities. Luxembourg (-26.18%) is the most extreme outlier, reflecting large λ increases in all three sectors. Germany (-12.55%), Denmark (-11.52%), and Austria (-10.69%) are next, driven primarily by services. Countries with a mix of rising and falling sectoral input intensities show partially offsetting gaps.

The Role of Input Intensity Variation. Table 1 suggests that the magnitude of the TFP gap is closely associated with the magnitude of the change in input intensity λ . Countries and sectors that experience larger shifts in λ tend to display larger differences between TFP measured with time-varying and constant input intensities. While this cross-sectional comparison is informative, it masks substantial year-to-year variation in sectoral input intensities within countries.

To examine whether this relationship holds systematically across countries, sectors, and over time, we turn to a panel regression framework. For each sector $n \in \{a, m, s\}$, we estimate

$$\Delta (\log A_n^i(t) - \log \bar{A}_n^i(t)) = \beta_n \Delta \log \lambda_n^i(t) + \gamma_i + \theta_t + \varepsilon_n^i(t), \quad (4.16)$$

where $A_n^i(t)$ denotes sector- n TFP constructed using time-varying input intensities and $\bar{A}_n^i(t)$ denotes TFP computed under constant λ in country i at time t . The dependent variable is the

¹¹Figures C.5–C.7 in Appendix C report the three components of the sectoral TFP growth gap: the value-added block gap, the intermediate-input block gap, and the input-intensity shift effect. The intermediate-input block gap is generally negligible across sectors and countries. In agriculture and manufacturing, the value-added and input-intensity shift components typically display opposite signs and largely offset each other over time. In contrast, services exhibit a different pattern: the value-added block gap is persistently negative (except for Luxembourg), while the input-intensity shift effect reinforces rather than offsets the value-added contribution.

change in the log TFP gap, while the regressor $\Delta \log \lambda_n^i(t)$ captures the growth rate of sectoral input intensity. Country fixed effects γ_i absorb time-invariant cross-country heterogeneity, and year fixed effects θ_t control for shocks common to all countries. Standard errors are clustered at the country level. This specification allows us to exploit the within-country time variation in input intensities and provides a transparent way to quantify whether periods of large adjustments in λ are systematically associated with larger movements in the TFP gap.

The regression in Equation (4.16) is not intended to establish a causal relationship. Its purpose is to characterize the heterogeneity in the sensitivity of measured TFP to input intensity changes across sectors and over time, and to test whether that sensitivity varies in ways consistent with the structural differences in production technology across agriculture, manufacturing, and services predicted by the decomposition in Section 3. The country and year fixed effects absorb time-invariant cross-country heterogeneity and common aggregate shocks, so the identifying variation is the within-country movement in sectoral input intensities. The regression coefficients therefore quantify the average within-country sensitivity of the TFP gap to a given change in input intensity for each sector, and differences in these coefficients across sectors carry economic content rooted in the structural differences in intermediate input shares documented in Section 2.

Table 2: TFP Divergence Panel Regression Results

	Agriculture		Manufacturing		Services	
	(1)	(2)	(1)	(2)	(1)	(2)
$\Delta \log \lambda_n^i(t)$	-0.379*** (0.022)	0.178** (0.061)	-0.261*** (0.032)	0.408*** (0.029)	-0.520*** (0.059)	-0.149*** (0.034)
Country FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Observations	336	336	336	336	336	312
Countries	14	14	14	14	14	13
R-squared	0.81	0.46	0.77	0.89	0.94	0.83

Notes: This table reports estimates of Equation (4.16). The dependent variable is the change in the log TFP gap, $\Delta(\log A_n^i(t) - \log \bar{A}_n^i(t))$. The independent variable is the growth rate of the sectoral input intensity, $\Delta \log \lambda_n^i(t)$. Column (1) uses the TFP gap computed with adjusted data; column (2) uses the gap computed without data adjustment under constant input intensity. Country and year fixed effects are included in all specifications. Standard errors are clustered at the country level and reported in brackets. Luxembourg is excluded from column (2) for services. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The estimates in Table 2 confirm and sharpen the patterns in Table 1. In column (1), which uses adjusted data, all three sectors display negative and highly significant coefficients: agriculture ($\hat{\beta}_a = -0.379$, $p < 0.01$), manufacturing ($\hat{\beta}_m = -0.261$, $p < 0.01$), and services ($\hat{\beta}_s = -0.520$, $p < 0.01$). A faster rise in λ is associated with a more negative TFP gap in every sector, equivalently, with greater overstatement by the constant-share approach.

The magnitude of the coefficients differs across sectors. Services has the largest coefficient in absolute value, indicating that services TFP is the most sensitive to movements in λ .

Agriculture is next, and manufacturing is the least sensitive.

Column (2) in Table 2, based on unadjusted data, provides a complementary view. For

agriculture ($\hat{\beta}_a = +0.178$, $p < 0.05$) and manufacturing ($\hat{\beta}_m = +0.408$, $p < 0.01$), the coefficient turns positive: without data adjustment, faster λ growth is associated with a more underestimation of productivity. For services ($\hat{\beta}_s = -0.149$, $p < 0.01$), the coefficient remains negative.¹² Interpreting this contrast correctly requires recognising that the full TFP gap in column (1) reflects three distinct channels, as formalised in the decomposition of Section 3.2 and derived in Appendix B.

The first is the value-added block channel: holding λ fixed affects how real value added is measured. When input intensity is relatively low, value added is effectively overstated, while it is understated when input intensity is high. This, in turn, carries over to measured value-added TFP. The second is the intermediate-input block channel: this reflects the contribution of intermediate inputs to productivity. Even when intermediate input quantities are unchanged, their role in measured gross output TFP varies with their share in production. The third is the input-intensity shift effect: changes in $\lambda_n(t)$ reallocate the relative importance of value added and intermediate inputs in production. As input intensity rises, more weight is placed on intermediate inputs and less on value added, thereby altering how sectoral intermediate input efficiency and value-added TFP contributions are reflected in measured gross output TFP. The magnitude of this effect depends on how large the change in input intensity is and on the relative scale of real value added to real intermediate inputs, as captured by $\log(V_n(t)/E_n(t))$.¹³

Under unadjusted data, in column (2), value added and intermediate inputs are taken directly from the observed series, so their implied productivity growth rates are the same in both measures. As shown in Equation (B.13) in Appendix B, the gap therefore no longer arises from separate adjustments to the value-added and intermediate blocks. Instead, it reflects the remaining terms in the decomposition. The sign of the effect differs across sectors. When λ rises, the gap is negative in services but positive in agriculture and manufacturing. In goods sectors, this effect works against the other forces operating in column (1), partially offsetting the net overstatement¹⁴. This explains why column (2) coefficients are positive for agriculture and manufacturing while column (1) coefficients remain negative for all sectors.

This sectoral heterogeneity reflects systematic differences in log-odds ratio $\log(\lambda_n/(1 - \lambda_n))$ adjusted for the relative price of value added versus intermediate inputs across sectors. In services, value added represents a large share of production ($\lambda_s < 0.5$ for most countries except Luxembourg¹⁵ in our panel) and its relative price tends to be low compared with intermediate inputs (See Figure A.2 in Appendix A). As a result, the term inside the logarithm is typically below one, so ϕ_n is negative. When the intermediate share λ_n rises, the shift effect therefore pushes measured productivity downward.

In goods sectors such as agriculture and manufacturing, intermediate inputs represent a larger share of production (above 0.5) and their prices play a relatively lower role. In this case the term

¹²We exclude Luxembourg from the services estimation reported in column (2), as—unlike other countries in the panel—services gross output in Luxembourg is more heavily driven by intermediate inputs.

¹³The relative scale of real value added to real intermediate inputs depends on the level of intermediate input intensity as well as on the relative price index between value added and intermediate inputs.

¹⁴See Figures C.5–C.7 in Appendix C, which plot the three contributions to the difference in TFP growth gaps under the two measures reported in column (1) of Table 2.

¹⁵See Figure A.1 in Appendix A.

inside the logarithm is typically above one, so ϕ_n is positive. An increase in λ_n then generates the opposite effect on measured productivity. This systematic difference in production structure explains why the input-intensity shift effect produces negative gaps in services but positive gaps in goods sectors when intermediate input intensities rise.

5 Discussion

The quantitative exercise in Section 4 should be read as deliberately conservative. The analysis is restricted to 14 advanced economies over 1995–2020, reflecting the availability of harmonized data on sectoral capital services, labor compensation, and inputs in EUKLEMS. This constraint limits the variation in input intensities within the sample, which remains modest for most countries, with the notable exception of Luxembourg.

This feature is quantitatively consequential. As documented in Section 2, input-intensity variations can be substantially larger over longer horizons. The mechanism we emphasize therefore operates over a relatively compressed range in our baseline sample, which mechanically attenuates its effect on measured TFP. The gaps reported in Table 1 should thus be interpreted as lower bounds on the quantitative importance of accounting for input-intensity dynamics. For illustration, the WORLD KLEMS database shows that the United States experienced an increase of more than 70% in intermediate input intensity in agriculture between 1947 and 1995.

A related concern pertains to the generalizability of our findings. The lack of consistent sectoral capital data precludes the implementation of the full quantitative framework for emerging and developing economies, thereby leaving open the extent to which the results extend beyond the set of advanced economies considered here. However, the stylized facts in Section 2, based on the WIOD long-run database and covering a broader set of countries—including India, China, Mexico, and South Korea—are informative in this respect. Input-intensity dynamics in these economies are at least as pronounced as in the baseline sample, and in several cases markedly stronger, suggesting that the effects identified for advanced economies may extend to emerging and developing countries.

Another natural question is whether the distinction between time-varying and constant-share TFP measures quantitatively matters for the interpretation of structural transformation. One of the traditional mechanism emphasized in the literature, often referred to as the Baumol effect, posits that labor reallocates toward sectors with relatively lower productivity growth. In this framework, slower productivity growth raises relative prices in those sectors, increases their expenditure shares, and thereby attracts resources. To the extent that differential productivity growth across sectors is a key driver of structural transformation, any heterogeneity in the measurement of sectoral productivity growth could, in principle, affect the inferred magnitude of these differentials and, consequently, the quantitative interpretation of structural transformation.

In particular, since the magnitude of TFP bias differs across sectors, the implied dynamics of relative productivity, such as A_a/A_m and A_s/A_m , under constant input-intensity assumptions would generally deviate from their counterparts based on time-varying measures. Figure 6 suggests a systematic pattern across most countries: both ratios tend to be somewhat overestimated under the constant-share assumption relative to the time-varying benchmark. This, in

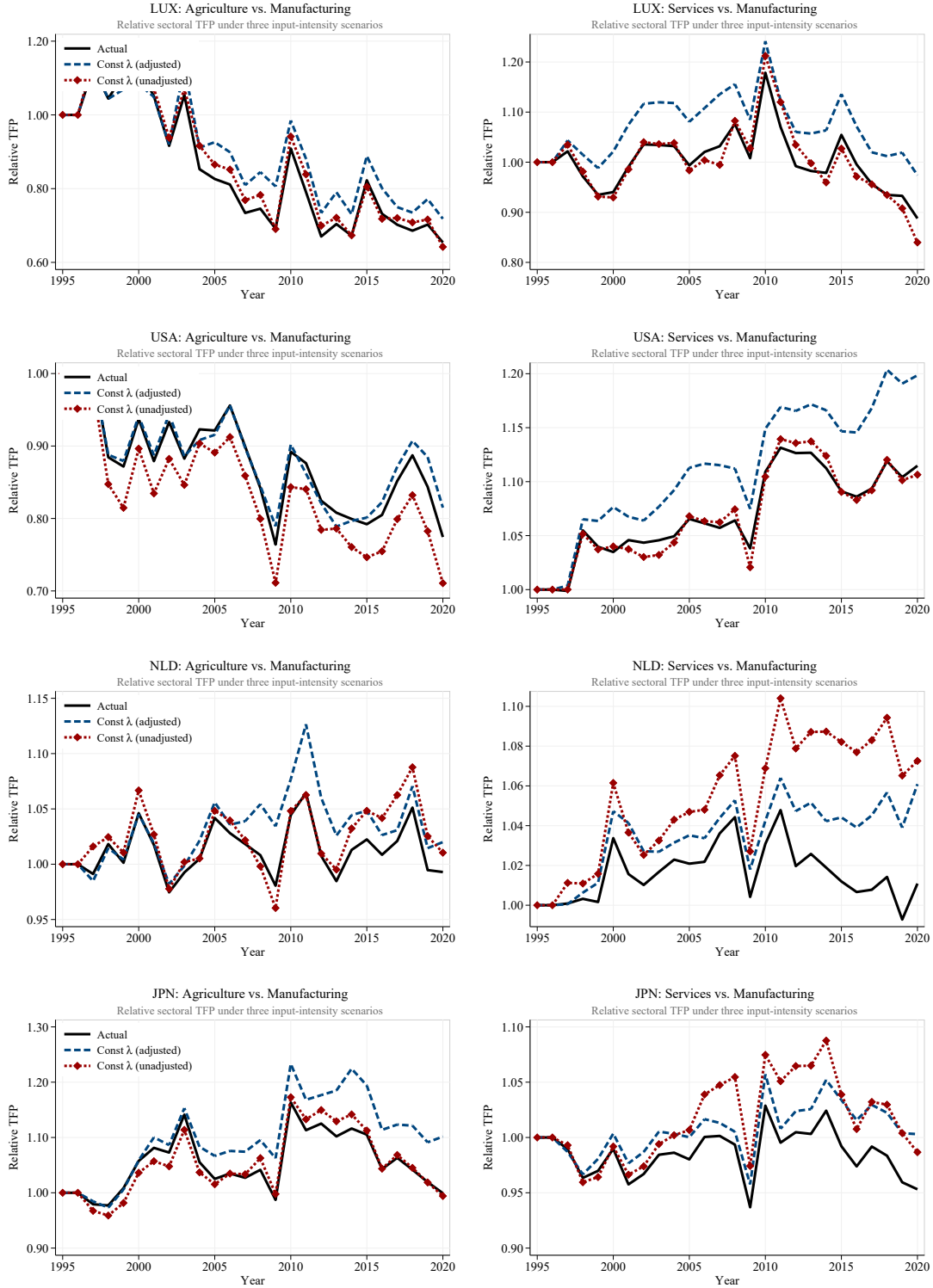


Figure 6: Baumol Effect: Relative Sectoral TFP Dynamics

Notes: This figure plots the TFP of agriculture (left column) and services (right column) relative to manufacturing, indexed to 1 in 1995, for Luxembourg (LUX), the United States (USA), the Netherlands (NLD), and Japan (JPN). The solid black line uses time-varying intermediate input intensities (actual). The dashed dark blue line uses constant input intensities with data adjusted to maintain national accounts consistency. The short-dashed dark red line with diamond markers uses constant input intensities without data adjustment, replicating the standard approach in the structural transformation literature. Divergence between the solid black line and the two constant-intensity lines reflects the Baumol-relevant measurement gap introduced by treating input intensities as fixed. Results for all remaining countries are in Figures C.8–C.10.

turn, could lead to an overstatement of the reallocation incentive from agriculture to manufacturing and an understatement of the reallocation from manufacturing to services, relative to the time-varying benchmark.

In a quantitative multi-sector model, calibrations based on constant-share TFP could therefore potentially imply a stronger reallocation of labor from agriculture to manufacturing and a weaker reallocation from manufacturing to services.

6 Conclusion

Measuring productivity in gross-output or input-output frameworks requires accounting for intermediate inputs explicitly. The standard approach assumes that intermediate input intensity is constant over time. This paper shows that this assumption affects the measured TFP growth.

National accounts link nominal gross output, value added, and intermediate expenditures. When intermediate input intensities vary over time, imposing a constant share breaks this accounting consistency because the ratio of intermediates to output is observed to change. Since gross output and intermediate expenditures are taken from the data, the discrepancy is absorbed by measured value added, which must be rescaled to preserve the national accounts identity. The value-added productivity residual can therefore differ depending on how intermediate input intensity is treated. When input intensities rise, a constant-share specification attributes an excessively large portion of output to value added, leading to an overstatement of value-added TFP and its contribution to gross-output TFP, and a corresponding understatement of the contribution of intermediate input efficiency.

Empirically, the differences between the time-varying and constant-share approaches are non-negligible. Across 14 advanced economies over 1995–2020, the gap in aggregate TFP between the two measures often reaches double digits. For example, Luxembourg’s aggregate TFP is about 26 percentage points higher under the constant-share approach and Germany’s by nearly 13, while Latvia’s is about 8 percentage points lower. Because sectoral productivity differentials play a central role in the reallocation of labor across sectors, differences in measured productivity across sectors may also affect the interpretation of structural transformation. The evidence therefore suggests that the evolution of production structure, captured by changing intermediate input intensities, must be incorporated into multisector growth models. As global value chains expand and production networks deepen, assuming fixed input shares becomes increasingly restrictive. The framework developed here provides a tractable way to account for these changes and to isolate the sources of mismeasurement, opening the door to future work linking shifts in input intensities to trade integration, technological change, and market structure.

A natural limitation of the present framework is that it does not explain why intermediate input intensities change over time. We treat their evolution as a measurable primitive rather than as the outcome of a structural model, which is what allows the accounting decomposition to hold exactly across all countries and sectors, but it leaves open the question of what drives the heterogeneous dynamics documented in Section 2. Several candidate mechanisms deserve investigation in future work, including the expansion of global value chains, the rising tradability of services, and factor-biased technological change. The framework developed here provides a

natural starting point: by combining our decomposition with a structural model that endogenizes input intensity dynamics through one or more of these channels, future work could quantify how much of the measured TFP gap is attributable to trade integration, technological change, or other structural forces.

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A Data Appendix



Figure A.1: Evolution of Sectoral Intermediate Input Shares Across Countries (EUKLEMS Data).

Notes: This figure plots the evolution of intermediate input shares (λ_{nt}) in agriculture, manufacturing, and services over time. Agriculture is shown with a dashed green line, manufacturing with a solid red line, and services with blue markers.



Figure A.2: Evolution of Sectoral Relative Price of Value Added to Intermediate Inputs.

Notes: This figure plots the evolution of the relative price of value added to intermediate inputs across agriculture, manufacturing, and services over time. Agriculture is shown with a dashed green line, manufacturing with a solid red line, and services with blue markers.

B Mathematical Appendix

This appendix derives the expressions for value-added TFP, intermediate-input TFP, and the TFP gap between the time-varying and constant-intensity scenarios. All derivations proceed in continuous time. Dots denote time derivatives, and $\bar{\lambda}_n$ denotes the sample average of $\lambda_n(t)$, which is held fixed in the constant-intensity scenario.

B.1 Proof

Proof of Equation (3.12). Optimal conditions imply that the cost share of input k in the intermediate bundle in sector n is given by

$$s_{E,nk}(t) \equiv \frac{P_k(t)E_{nk}(t)}{P_{E,n}(t)E_n(t)} \quad (\text{B.1})$$

where the denominator is the total intermediate expenditure in the sector n , consequently

$$P_{E,n}(t)E_n(t) = \sum_{k \in \{a,m,s\}} P_k(t)E_{nk}(t) \quad (\text{B.2})$$

Thanks to CES aggregator properties, the cost share $s_{E,nk}$ can be rewritten as

$$s_{E,nk}(t) = \frac{\omega_{E,nk}^{\varepsilon_n} P_k(t)^{1-\varepsilon_n}}{\sum_{j \in \{a,m,s\}} \omega_{E,nj}^{\varepsilon_n} P_j(t)^{1-\varepsilon_n}} \quad (\text{B.3})$$

Taking a total differential of the Equation (B.2) and divide by $P_{E,n}(t)E_n(t)$ yields

$$\frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} + \frac{\dot{E}_n(t)}{E_n(t)} = \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \left(\frac{\dot{P}_k(t)}{P_k(t)} + \frac{\dot{E}_{nk}(t)}{E_{nk}(t)} \right) \quad (\text{B.4})$$

Similarly, total differentiation of the logarithm of Equation (3.5) and rearranging yields

$$\frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} = \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \frac{\dot{P}_k(t)}{P_k(t)} - \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} \quad (\text{B.5})$$

Combining these yields exactly

$$\frac{\dot{E}_n(t)}{E_n(t)} = \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} + \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \frac{\dot{E}_{nk}(t)}{E_{nk}(t)}, \quad (\text{B.6})$$

B.2 Value-Added Block TFP

Setup. Recall from Section 3.1 that sector n produces gross output $Y_n(t)$ using capital, labor, and intermediate inputs. Real value added is defined as the portion of gross output not absorbed

by intermediate inputs, deflated by the value-added price index:

$$V_n(t) = \frac{(1 - \lambda_n(t)) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_{V,n}(t)},$$

where $\lambda_n(t) = P_{E,n}(t)E_n(t)/[P_n(t)Y_n(t)]$ is the intermediate input intensity. Value-added TFP, $A_{V,n}(t)$, is defined as the Solow residual from the value-added production function:

$$\frac{\dot{V}_n(t)}{V_n(t)} = \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} + \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} + (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)}.$$

Growth Rate of Real Value Added. Differentiating the expression for $V_n(t)$ with respect to time gives:

$$\frac{\dot{V}_n(t)}{V_n(t)} = \frac{\dot{\lambda}_n(t)}{1 - \lambda_n(t)} + \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{V,n}(t)}{P_{V,n}(t)}.$$

The term $\dot{\lambda}_n(t)/(1 - \lambda_n(t))$ appears because the value-added share $(1 - \lambda_n(t))$ itself changes over time. It is absent in the constant-intensity case, which is the source of mismeasurement.

Value-Added TFP Growth. Substituting into the definition of $A_{V,n}(t)$:

$$\begin{aligned} \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} &= \frac{\dot{V}_n(t)}{V_n(t)} - \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} - (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)} \\ &= \frac{\dot{\lambda}_n(t)}{1 - \lambda_n(t)} + \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{V,n}(t)}{P_{V,n}(t)} - \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} - (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)}. \end{aligned}$$

Under a constant $\lambda_n = \bar{\lambda}_n$, the $\dot{\lambda}_n(t)$ term vanishes:

$$\frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} = \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{V,n}(t)}{P_{V,n}(t)} - \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} - (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)}.$$

The Value-Added TFP Gap. Subtracting the constant-intensity expression from the time-varying one, after weighting by their respective value-added shares:

$$\begin{aligned} (1 - \lambda_n(t)) \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} - (1 - \bar{\lambda}_n) \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} &= -\dot{\lambda}_n(t) - (\lambda_n(t) - \bar{\lambda}_n) \left[\frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{V,n}(t)}{P_{V,n}(t)} \right. \\ &\quad \left. - \alpha_n \frac{\dot{K}_n(t)}{K_n(t)} - (1 - \alpha_n) \frac{\dot{L}_n(t)}{L_n(t)} \right]. \end{aligned}$$

The first term, $-\dot{\lambda}_n(t)$, captures the direct effect of a changing value-added share. The second term captures the input-intensity shift effect: whenever $\lambda_n(t) \neq \bar{\lambda}_n$, the two scenarios apply different weights to the same gross-output growth rate, generating an additional wedge.

B.3 Intermediate-Input Block TFP

Setup. Total real intermediate inputs $E_n(t)$ and their sector- k components $E_{nk}(t)$ are constructed from observed nominal expenditures deflated by the appropriate price indices:

$$E_n(t) = \frac{\lambda_n(t) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_{E,n}(t)}, \quad E_{nk}(t) = \frac{\widetilde{s}_{E,nk}(t) \lambda_n(t) \widetilde{P}_n(t) \widetilde{Y}_n(t)}{\widetilde{P}_k(t)}, \quad (\text{B.7})$$

where $s_{E,nk}(t)$ is the nominal expenditure share of sector- k inputs in sector n 's total intermediate use. Intermediate-input TFP, $A_{E,n}(t)$, is defined as the Solow residual from a CES aggregator over these inputs:

$$\frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} = \frac{\dot{E}_n(t)}{E_n(t)} - \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \frac{\dot{E}_{nk}(t)}{E_{nk}(t)}. \quad (\text{B.8})$$

Growth Rates of Intermediate Aggregates. Differentiating aggregate intermediate $E_n(t)$:

$$\frac{\dot{E}_n(t)}{E_n(t)} = \frac{\dot{\lambda}_n(t)}{\lambda_n(t)} + \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)}, \quad (\text{B.9})$$

$$\frac{\dot{E}_{nk}(t)}{E_{nk}(t)} = \frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)} + \frac{\dot{\lambda}_n(t)}{\lambda_n(t)} + \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_k(t)}{P_k(t)}. \quad (\text{B.10})$$

Note that $\dot{\lambda}_n(t)/\lambda_n(t)$ enters symmetrically into both $\dot{E}_n/E_n$ and $\dot{E}_{nk}/E_{nk}$, so it cancels out in the Solow residual. Intermediate-input TFP growth is therefore independent of $\lambda_n(t)$ conditional on prices and expenditure shares. The growth rate of intermediate-input productivity can be written as

$$\frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} = \sum_k s_{E,nk}(t) \frac{\dot{P}_k(t)}{P_k(t)} - \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} - \sum_k s_{E,nk}(t) \frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)}. \quad (\text{B.11})$$

This expression highlights two distinct sources of intermediate-input efficiency dynamics:

$$\frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} = \underbrace{\left(\sum_k s_{E,nk}(t) \frac{\dot{P}_k(t)}{P_k(t)} - \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} \right)}_{\text{Relative supplier price dynamics}} - \underbrace{\sum_k s_{E,nk}(t) \frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)}}_{\text{Reallocation across sectoral linkages}}. \quad (\text{B.12})$$

The dynamics of intermediate-input productivity are governed by the sectoral composition of linkages and relative price movements across supplying industries. The expenditure shares $s_{E,nk}$ capture the strength of inter-sectoral connections and determine how efficiency gains in supplier sectors propagate to the downstream sector n . Changes in these shares reflect reallocation across suppliers. Increasing reliance on relatively cheaper or more productive intermediates raises measured intermediate-input efficiency growth, whereas shifts toward relatively more expensive or less productive inputs reduce it.

Similarly, the difference between the weighted growth of supplier prices and the growth of the aggregate intermediate price index captures the effect of relative price movements across sectors. Uniform price changes across suppliers leave this component unchanged, whereas differential price dynamics reflect productivity changes in upstream industries that transmit through the input-output network.

Intermediate-Input Efficiency Growth. Substituting and simplifying (the λ -growth terms cancel because $\sum_k s_{E,nk}(t) = 1$):

$$\begin{aligned}
\lambda_n(t) \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} &= \lambda_n(t) \frac{\dot{E}_n(t)}{E_n(t)} - \lambda_n(t) \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \frac{\dot{E}_{nk}(t)}{E_{nk}(t)} \\
&= \dot{\lambda}_n(t) + \lambda_n(t) \left[\frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} \right] \\
&\quad - \lambda_n(t) \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \left[\frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)} + \frac{\dot{\lambda}_n(t)}{\lambda_n(t)} + \frac{\dot{P}_n(t)}{P_n(t)} + \frac{\dot{Y}_n(t)}{Y_n(t)} - \frac{\dot{P}_k(t)}{P_k(t)} \right] \\
&= -\lambda_n(t) \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} - \lambda_n(t) \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \left[\frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)} - \frac{\dot{P}_k(t)}{P_k(t)} \right].
\end{aligned}$$

Under constant $\lambda_n = \bar{\lambda}_n$, replacing $\lambda_n(t)$ with $\bar{\lambda}_n$ throughout:

$$\bar{\lambda}_n \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} = -\bar{\lambda}_n \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} - \bar{\lambda}_n \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \left[\frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)} - \frac{\dot{P}_k(t)}{P_k(t)} \right].$$

Intermediate-Input Efficiency Gap. Subtracting:

$$\lambda_n(t) \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} - \bar{\lambda}_n \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} = -(\lambda_n(t) - \bar{\lambda}_n) \left\{ \frac{\dot{P}_{E,n}(t)}{P_{E,n}(t)} + \sum_{k \in \{a,m,s\}} s_{E,nk}(t) \left[\frac{\dot{s}_{E,nk}(t)}{s_{E,nk}(t)} - \frac{\dot{P}_k(t)}{P_k(t)} \right] \right\}.$$

The gap in the intermediate block is proportional to $\lambda_n(t) - \bar{\lambda}_n$, weighted by the growth rate of the intermediate price index. This term is generally small relative to the value-added block gap, which explains why the dominant channel of mismeasurement operates through value added.

B.4 Sectoral Gross-Output TFP Gap

Sectoral gross-output TFP is a weighted average of the two blocks plus an input-intensity shift term (see Section 3.2). The gap between the time-varying and constant-intensity measures is:

$$\begin{aligned}
\frac{\dot{A}_n(t)}{A_n(t)} - \frac{\dot{\bar{A}}_n(t)}{\bar{A}_n(t)} &= \underbrace{(1 - \lambda_n(t)) \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} - (1 - \bar{\lambda}_n) \frac{\dot{\bar{A}}_{V,n}(t)}{\bar{A}_{V,n}(t)}}_{\text{value-added block gap}} \\
&\quad + \underbrace{\lambda_n(t) \frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} - \bar{\lambda}_n \frac{\dot{\bar{A}}_{E,n}(t)}{\bar{A}_{E,n}(t)}}_{\text{intermediate-input block gap}} \\
&\quad + \underbrace{\dot{\lambda}_n(t) \log \left(\frac{\lambda_n(t)}{1 - \lambda_n(t)} \frac{P_{V,n}(t)}{P_{E,n}(t)} \right)}_{\text{input-intensity shift effect}}.
\end{aligned}$$

The first two terms are derived in Sections B.2 and B.3 above. The third term is the input-intensity shift effect: it arises because a change in $\lambda_n(t)$ shifts the relative weight placed on the intermediate and value-added blocks, and the magnitude of this shift depends on the log-odds

ratio of $\lambda_n(t)$ adjusted for the relative price of value added versus intermediate inputs. When $\lambda_n(t)$ is constant, $\dot{\lambda}_n(t) = 0$ and this term vanishes exactly, leaving no contribution from this effect.

If intermediate input intensities are held constant, but value added and intermediate inputs are used without adjusting them to be consistent with gross output, the difference in productivity growth between the two measures is given by

$$\frac{\dot{A}_n(t)}{A_n(t)} - \frac{\dot{\bar{A}}_n(t)}{\bar{A}_n(t)} = (\lambda_n(t) - \bar{\lambda}_n) \left(\frac{\dot{A}_{E,n}(t)}{A_{E,n}(t)} - \frac{\dot{A}_{V,n}(t)}{A_{V,n}(t)} \right) + \dot{\lambda}_n(t) \log \left(\frac{\lambda_n(t)}{1 - \lambda_n(t)} \frac{P_{V,n}(t)}{P_{E,n}(t)} \right). \quad (\text{B.13})$$

C Result Appendix

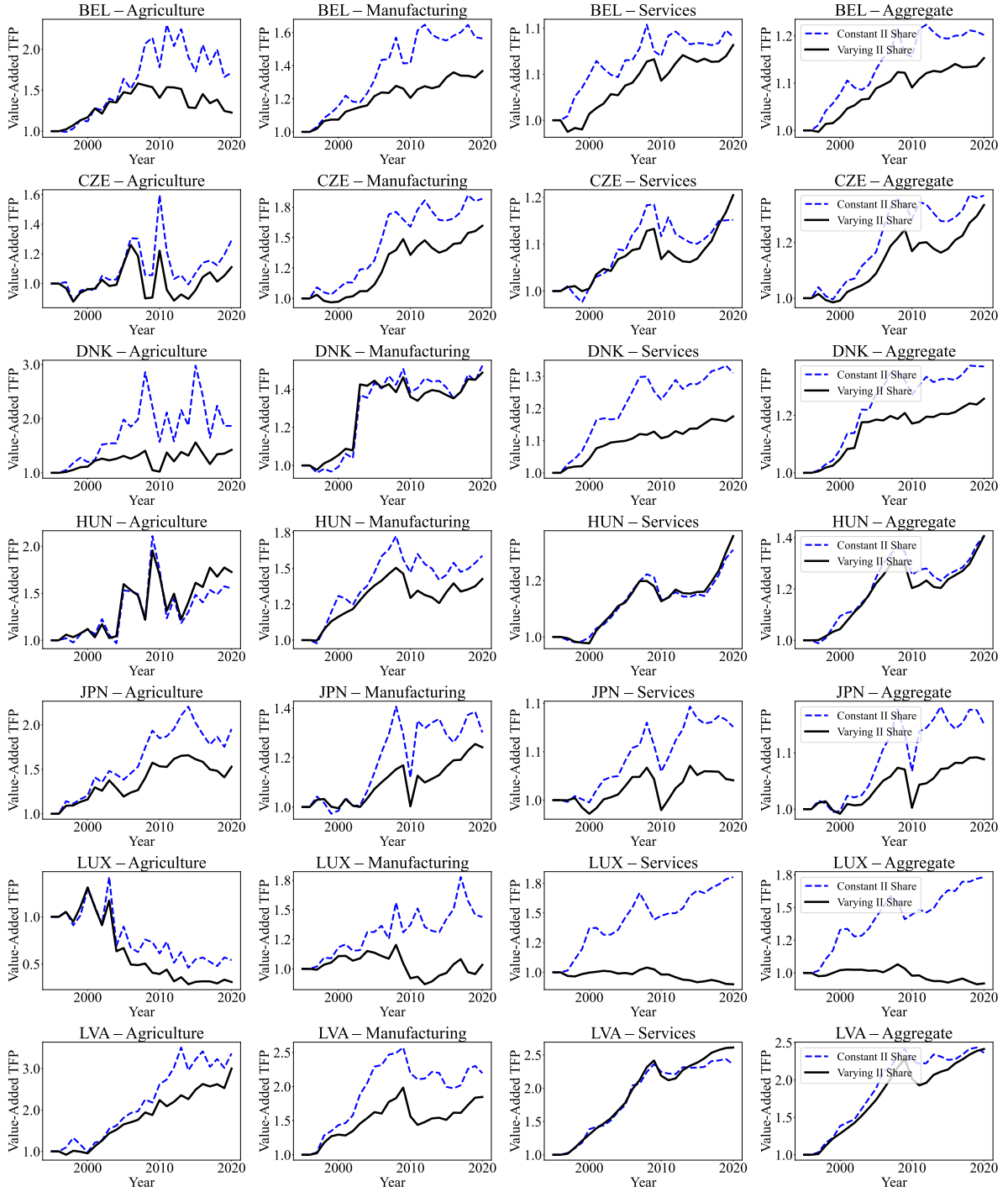


Figure C.3: Sectoral and Aggregate Value-Added TFP Trends – Panel 1

Notes: This figure displays sectoral and aggregate value-added TFP indices (1995=100) over time for Belgium (BEL), Czech Republic (CZE), Denmark (DNK), Hungary (HUN), Japan (JPN), Luxembourg (LUX), and Latvia (LVA). Each row shows agriculture, manufacturing, services, and aggregate TFP from left to right. The solid black line is value-added TFP computed with time-varying intermediate input intensities. The dashed blue line is TFP computed with constant input intensities and adjusted data.

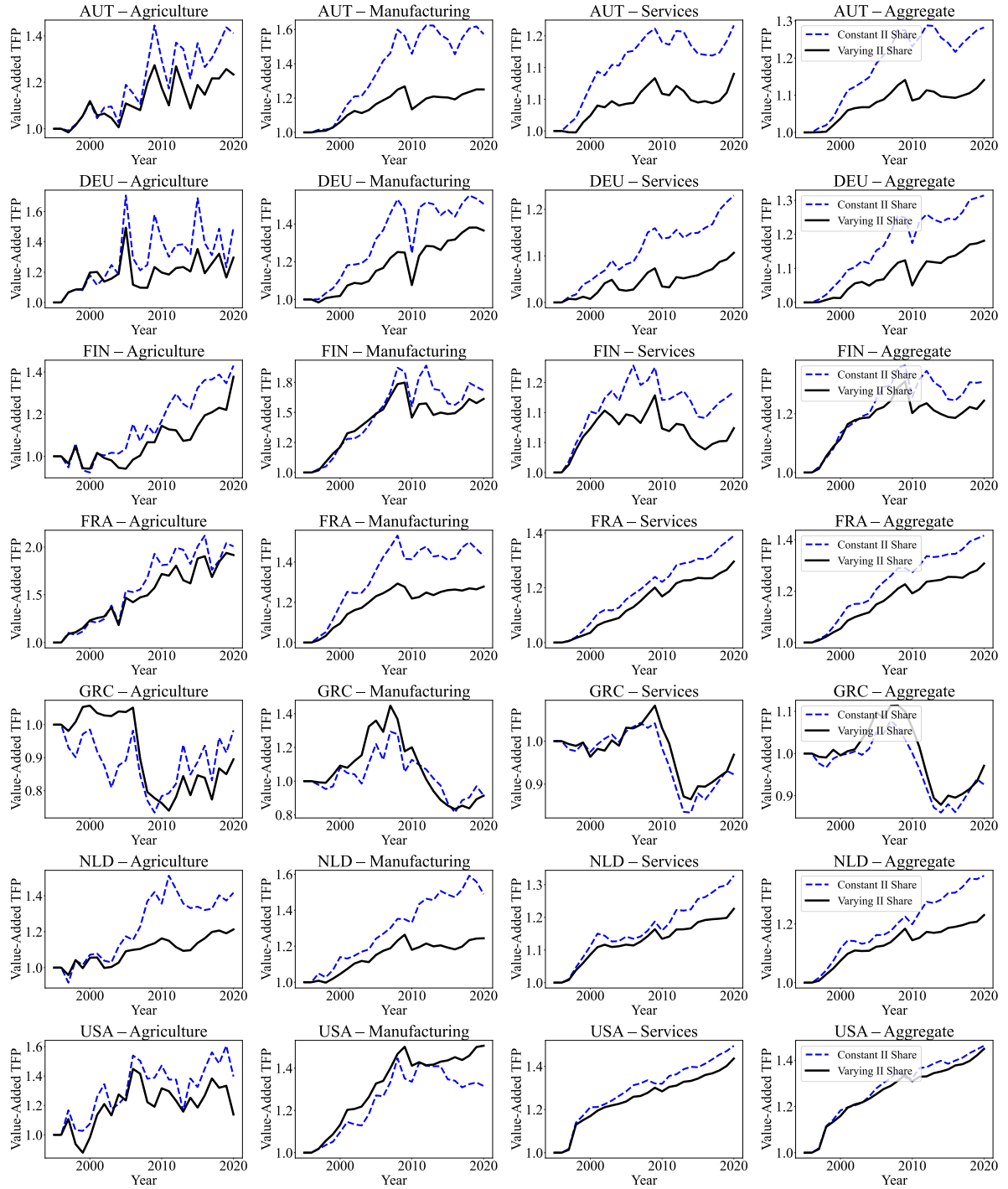


Figure C.4: Sectoral and Aggregate Value-Added TFP Trends – Panel 2

Notes: This figure displays sectoral and aggregate value-added TFP indices (1995=100) over time for Austria (AUT), Germany (DEU), Finland (FIN), France (FRA), Greece (GRC), Netherlands (NLD), and United States (USA). Each row shows agriculture, manufacturing, services, and aggregate TFP from left to right. The solid black line is value-added TFP computed with time-varying intermediate input intensities. The dashed blue line is value-added TFP computed with constant input intensities and adjusted data.

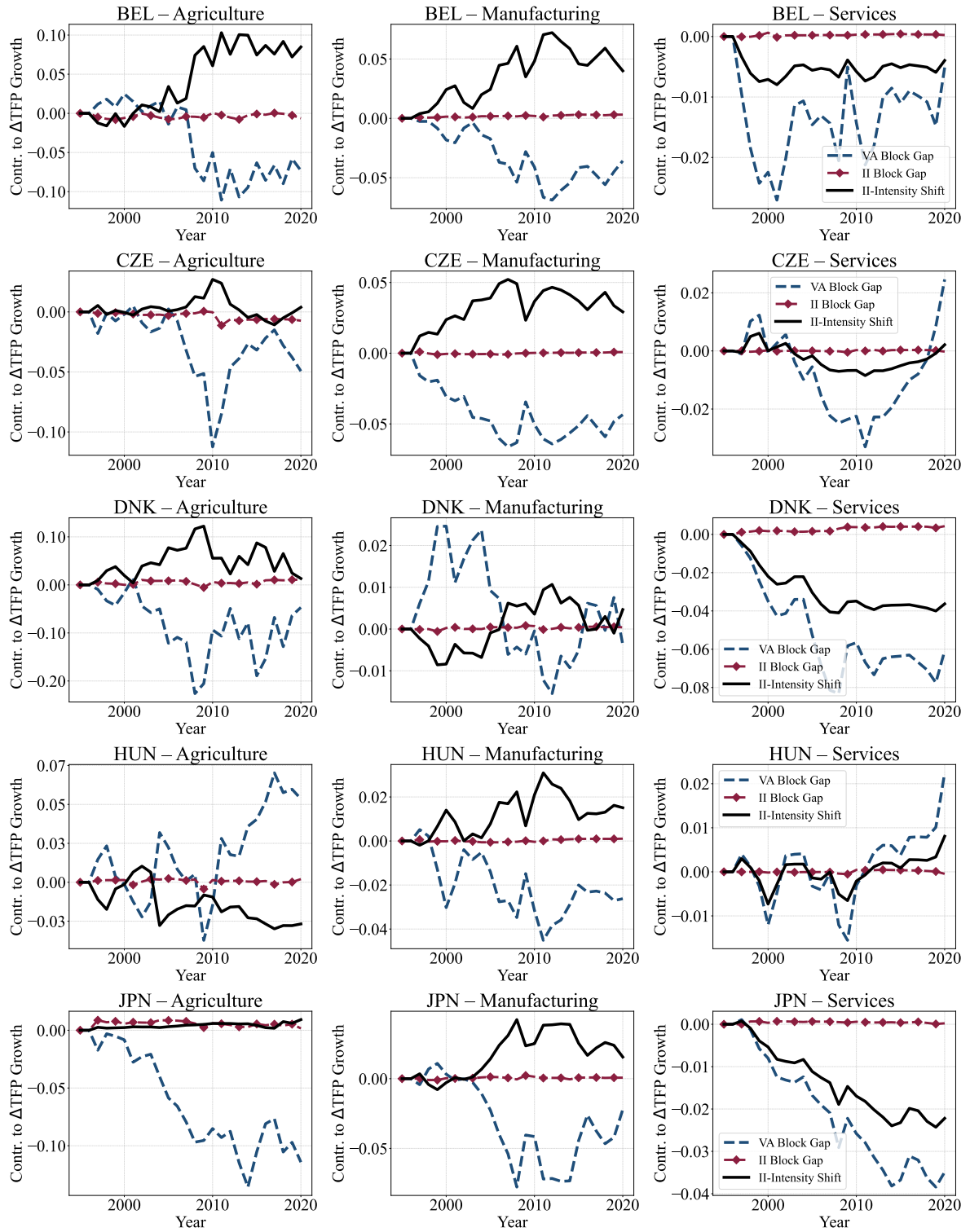


Figure C.5: Contributions to TFP Growth Gaps – Panel 1

Notes: This figure plots the three components of the productivity-growth gap from Appendix B.4 for Belgium (BEL), Czech Republic (CZE), Denmark (DNK), Hungary (HUN), and Japan (JPN). Each row shows agriculture, manufacturing, and services from left to right. The solid black line is the input-intensity shift effect. The dashed blue line is the value-added block gap. The dashed-marked red line is the intermediate-input block gap.

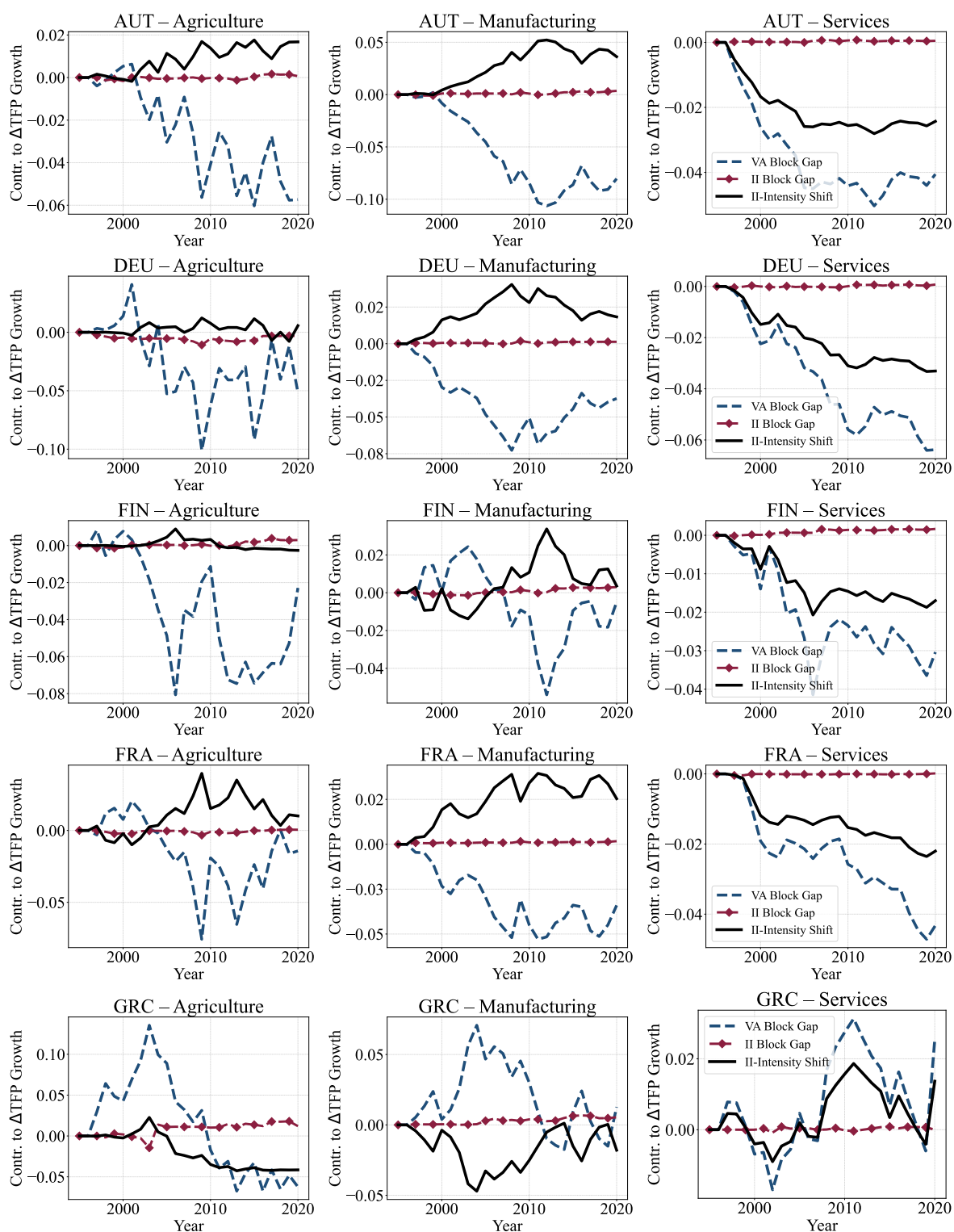


Figure C.6: Contributions to TFP Growth Gaps – Panel 2

Notes: This figure plots the three components of the productivity-growth gap from Appendix B.4 for Austria (AUT), Germany (DEU), Finland (FIN), France (FRA), and Greece (GRC). Each row shows agriculture, manufacturing, and services from left to right. The solid black line is the input-intensity shift effect. The dashed blue line is the value-added block gap. The dashed-marked red line is the intermediate-input block gap.

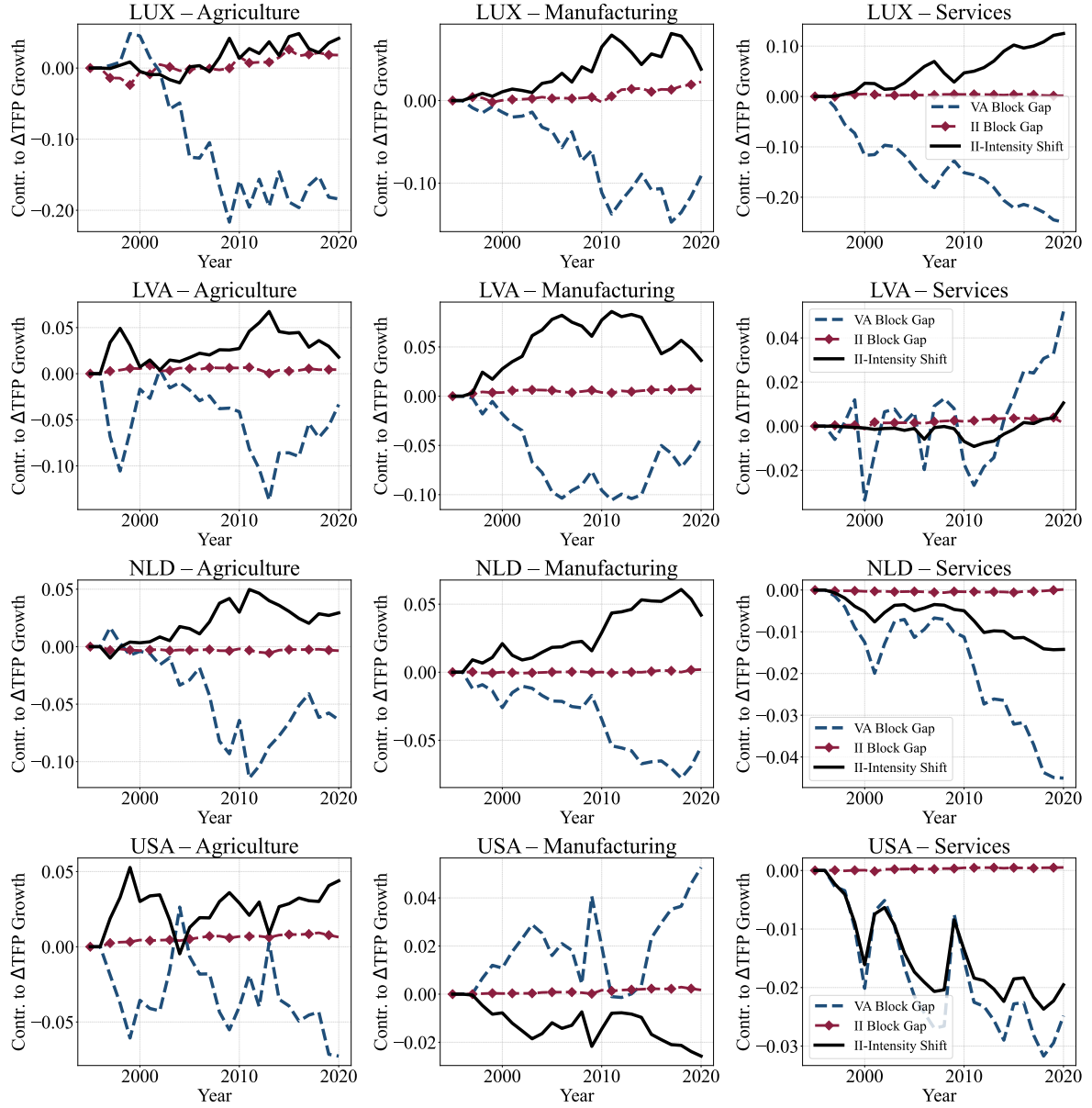


Figure C.7: Contributions to TFP Growth Gaps – Panel 3

Notes: This figure plots the three components of the productivity-growth gap from Appendix B.4 for Luxembourg (LUX), Latvia (LVA), Netherlands (NLD), and United States (USA). Each row shows agriculture, manufacturing, and services from left to right. The solid black line is the input-intensity shift effect. The dashed blue line is the value-added block gap. The dashed-marked red line is the intermediate-input block gap.

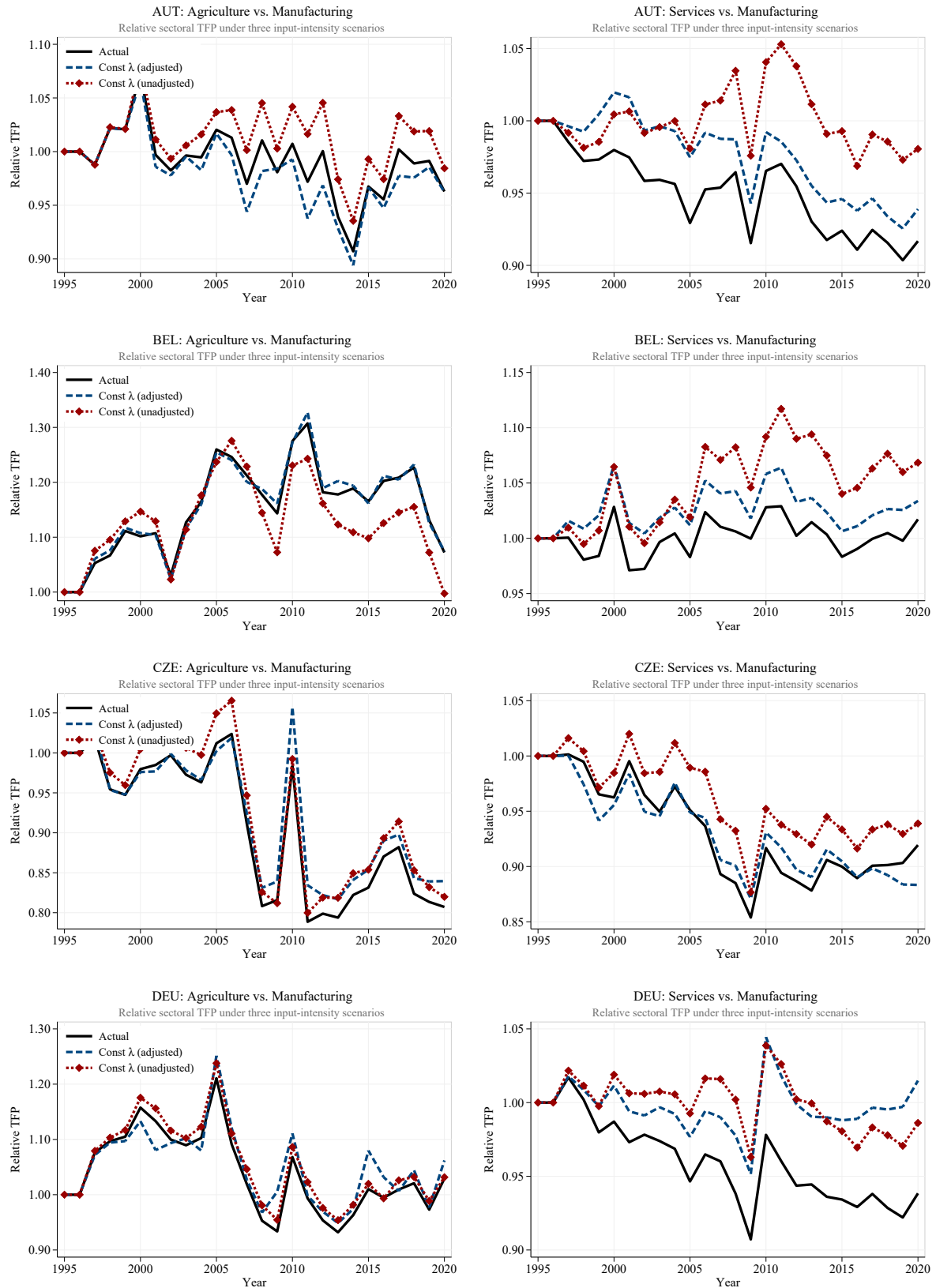


Figure C.8: Baumol Effect: Relative Sectoral TFP Dynamics – Appendix Panel 1

Notes: See Figure 6 for variable definitions. Countries shown: Austria (AUT), Belgium (BEL), Czech Republic (CZE), and Germany (DEU).

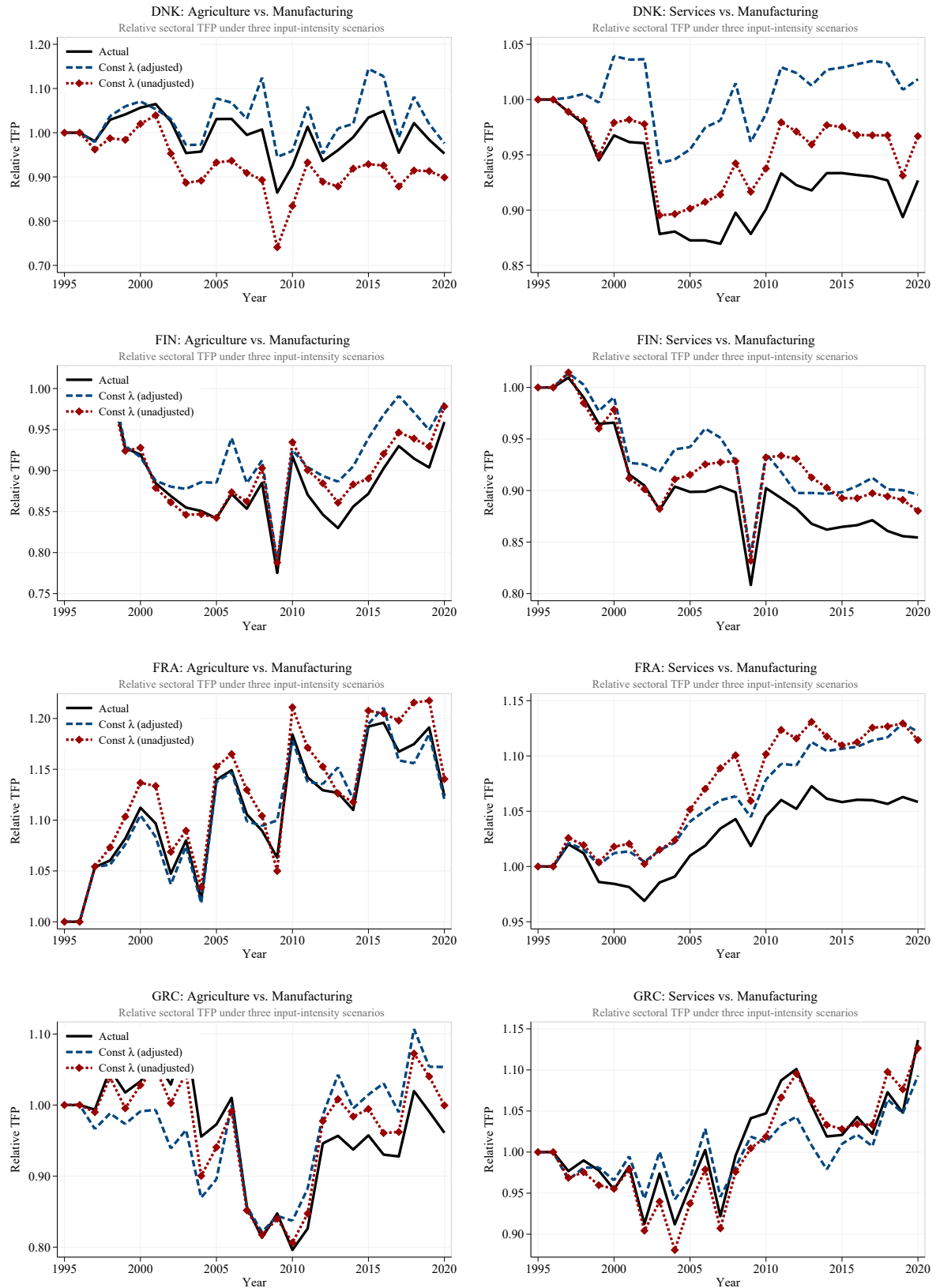


Figure C.9: Baumol Effect: Relative Sectoral TFP Dynamics – Appendix Panel 2

Notes: See Figure 6 for variable definitions. Countries shown: Denmark (DNK), Finland (FIN), France (FRA), and Greece (GRC).

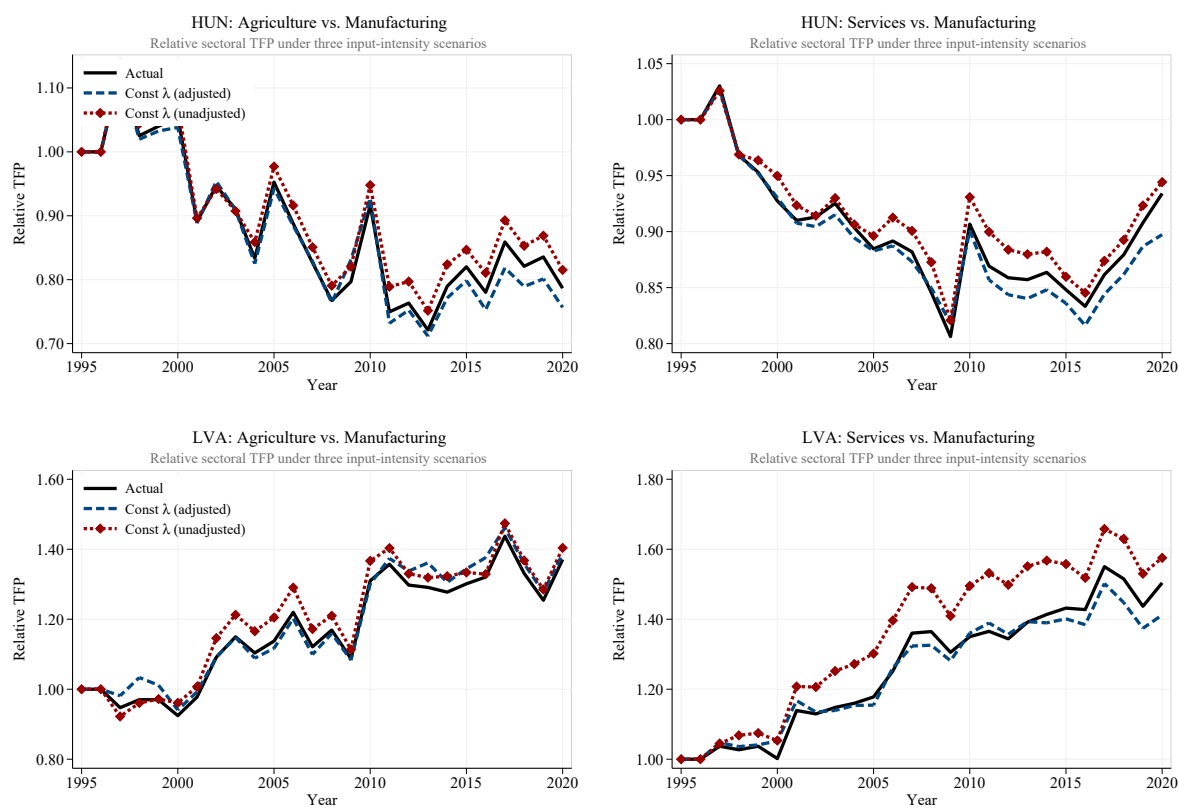


Figure C.10: Baumol Effect: Relative Sectoral TFP Dynamics – Appendix Panel 3

Notes: See Figure 6 for variable definitions. Countries shown: Hungary (HUN) and Latvia (LVA).